

Physics-Informed Residual Diffusion for UK Precipitation Nowcasting

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Abstract

Short-range rainfall forecasting relies on Lagrangian extrapolation, which uses recent radar data to determine a motion field and move the previous observation along it. Although skilled within an hour, it lacks the ability to expand or decay and loses its tiniest scales quickly. Recent generative nowcasting systems divide the problem into a deterministic and stochastic component. However, the deterministic component is learned on the same data as the generator, potentially leading to overfitting. This experiment explores what occurs when a part has no parameters at all. Pysteps are used to generate a physics-only advection prior A . A conditional latent diffusion model learns only the residual r left behind by advection. The nowcast is calculated as $A + \hat{r}$ and is compared to observed radar, not the model's training residual. Two arms were trained on 613,892 lead-conditioned samples of Met Office 1 km UK radar for one hour: direct residual diffusion and a CorrDiff-style version that first fits a frozen conditional-mean network. On 53,124 validation crops, both arms outperformed advection, with the heavy-rain critical success index increasing from 0.086 to 0.144. The two-stage arm consistently outperforms in all accuracy categories, with a margin of 2.8 to 8.2 percent increasing with rain rate. The ranking is stronger still on the held-out 2026 test year, where the heavy-rain margin reaches 17.8 percent. Averaging the ensemble reduces real-world resolution as a single member can perform well at 21 km but the mean requires 51 km.

Keywords: precipitation nowcasting, latent diffusion models, residual learning, weather radar, forecast verification, ensemble calibration.

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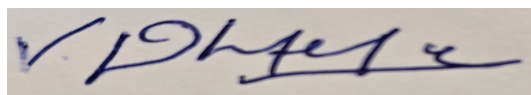
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1 Introduction

Forecasts of rainfall in the next 0 to 2 hours, known as nowcasts, fill an inconvenient gap. Numerical weather prediction takes longer to generate convection and is often outperformed by simpler radar-driven algorithms (Sun et al., 2014; Prudden et al., 2020). Forecasts inform immediate and physical decisions, such as closing roads, holding aircraft, opening flood gates, and moving people. The European nowcasting community anticipates that probabilistic, high-resolution, convection-scale guidance will be central to the next decade of these systems (Bojinski et al., 2023).

The method that has held the field for four decades is Lagrangian extrapolation, estimate a motion field from the last few radar frames, then transfer the most recent observation along it (Pulkkinen et al., 2019). It works because rain does move and most of the future field is the present field shifted within an hour. It concedes on a structural, rather than an accidental, ground. The scheme assumes the field is steady in a Lagrangian frame, so by construction it cannot grow a cell, decay one, or create a new one, and it loses small scales first because the semi-Lagrangian interpolation that puts displaced values back on a regular grid is a low-pass filter applied once per step.

Deep generative models offered a promising solution to the blurring problem. The Deep Generative Model of Radar (DGMR) generates sharp, spatiotemporally consistent ensembles up to 90 minutes on the same 1 km composite from the Met Office used here, and was chosen by operational meteorologists over an extrapolation method and a deterministic deep network (Ravuri et al., 2021). NowcastNet integrates a physically inspired evolution operator in the network and aims to precisely predict intense rainfall (Zhang et al., 2023). Diffusion models came in, first as full-field nowcasters (Gao et al., 2023; Leinonen et al., 2023; Dai et al., 2025), and then in a form directly applicable here: split the forecast into a deterministic component and a stochastic residual, and let the generative model learn only the residual (Yu et al., 2024; Mardani et al., 2025).

1.1 The gap this project addresses

It is simple to understand why the residual notion is so well-established. A residual is easier to target since its variance is smaller than that of the field from which it is extracted, and the split clearly separates labor between what is and is not predictable on average. However, the deterministic part of that divide is a learnt network in all published versions. DiffCast diffuses what it leaves behind after learning a deterministic motion backbone (Yu et al., 2024). In order to estimate the conditional mean, freeze it, and diffuse the remainder, CorrDiff learns a regression network (Mardani et al., 2025). The evolution operator of NowcastNet is taught end-to-end (Zhang et al., 2023). During joint training, the deterministic portion may drift, overfit, or be forgotten because it has parameters and is fitted to the same data as the generator. The conflict is evident in DiffCast’s own ablation, which lowers the critical success index from 0.3077 to 0.2739 by freezing its learned backbone instead of developing it simultaneously.

According to the literature search for this project, no one has posed the complementary question. What would happen if the deterministic component were a classical physical scheme with absolutely no free parameters? Since there is nothing to fit, such a component cannot overfit, and since it is never trained, it cannot be forgotten during training. In comparison to the diffusion sampler, it is less expensive, easier to understand, and its failure modes have already been reported in the meteorological literature. The residual approach becomes more appealing if a parameter-free deterministic stage is successful since the split no longer introduces a second potential problem.

There is one prior attempt in the neighbourhood, and it is a warning rather than a foundation. Chase et al. (2025) tried a latent two-stage nowcaster on satellite infrared brightness temperature, found its regression stage unhelpful, and abandoned it in a single sentence without reporting numbers. Their wider result is that off-the-shelf image autoencoders reconstruct meteorological fields poorly and inject small-scale noise, and their recommendation is to train a domain-specific codec. This project follows that recommendation and, as a secondary outcome, supplies the latent-regression result their paper leaves open.

1.2 Aim, research questions and contributions

The aim is to develop and evaluate a physics-informed residual diffusion nowcaster for UK rainfall at lead times of +15 to +60 minutes, and to judge it honestly against the physics it is built on.

The system is organised around one decomposition. Writing y for the observed radar field at the forecast lead, A for a non-learned advection forecast of it, and r for what advection misses,

$$y = A + r, \tag{1}$$

the diffusion model learns only r , and the nowcast $A + \hat{r}$ is always scored against y . Three research questions follow.

RQ1 (physical prior). Does learning only the residual left by a non-learned advection forecast beat that advection baseline, and persistence, on UK radar?

RQ2 (residual decomposition). Given a fixed physical prior, does additionally forecasting the conditional mean of the residual with a frozen learned stage beat learning that residual directly?

RQ3 (sharpness against accuracy). What is lost when the ensemble is collapsed to its mean, and at what spatial scale is each output actually skilful?

RQ2 is the direct test of the gap identified above. DiffCast froze a learned backbone and lost skill. The prior used here has no parameters and cannot be unfrozen, so RQ2 asks whether a second stage, learned and then frozen, still earns its place on top of one that never needed freezing in the first place.

There are five contributions. (i) A complete, reproducible residual nowcasting pipeline on Met Office 1 km UK radar with a parameter-free physical prior, trained and evaluated at four lead times on 260,492 quality-filtered crops. (ii) The first quantified answer, to this project’s knowledge, to whether a frozen learned conditional-mean stage adds value on top of a parameter-free physical prior, run as a controlled single-change ablation with a precommitted checkpoint-selection rule. (iii) A quantified account of what ensemble averaging costs, giving the smallest skilful spatial scale for the mean and for a single member side by side. (iv) The latent-regression measurement that Chase et al. (2025) left open, on a purpose-trained radar codec. (v) A negative but useful hyperparameter result: 43 configurations were screened across nine parameters and the incumbent survived, and the multi-fidelity search reports its own rank-transfer failure.

2 Background and related work

2.1 The extrapolation baseline and why it is hard to beat

Radar-based nowcasting starts from optical flow. A motion field is estimated from a short sequence of radar images, most commonly by a local least-squares tracker that solves the brightness-constancy constraint over a small window around each pixel, and the last frame is then transported along that field by a semi-Lagrangian scheme, which traces each destination pixel back along the flow to where its value came from and interpolates once. The open-source pysteps library packages both steps and the probabilistic extensions built on them (Pulkkinen et al., 2019; Bowler et al., 2006), and is the reference implementation used here.

Everything that follows is determined by two aspects of this baseline. It is very strong at short range, the future field is indeed the current field displaced out in time at +15 minutes, and it decays in a predictable fashion with lead time as growth and decay build. That decay is the opening a learnt model has, and it means any comparison against advection must be made lead by lead rather than pooled else the outcome is an average spanning regimes where the baseline is unbeatable and regimes where it is weak. The second property is that the interpolation step damps small scales. This project evaluates that damping directly: in the 2 to 8 km band, the advection field preserves 49 to 56 percent of observed spatial power depending on lead (Section 6.1).

2.2 Deep generative nowcasting

Early deep nowcasters were deterministic sequence predictors trained with a pixel-wise loss (Shi et al., 2015; Ayzel et al., 2020). They reduced mean error but produced increasingly blurred fields at longer leads, for a reason that is fundamental rather than architectural: the minimiser of a pixel-wise expected loss is a conditional mean, and the conditional mean of a sharp, displaced field is a smooth one. The same lesson has been drawn from the opposite direction by Metzl et al. (2025), who show that what a convolutional nowcaster can represent is set by whether its receptive field spans the physical scale of the motion it has to follow, and that supplying advection explicitly relieves the network of learning transport it is poorly shaped to learn. That is the architectural argument for the decomposition this project tests.

DGMR broke the pattern by training a conditional generative adversarial network with spatial and temporal discriminators and a grid-cell regulariser weighted by rain intensity (Ravuri et al., 2021). It is important in this context for three reasons other than the outcome. It uses the same Met Office 1 km UK composite in 256 km crops, making it directly similar to this one. The study found that comparing generative nowcasters to pysteps is more beneficial than comparing them to other neural networks. It explicitly stated that pointwise correctness and physical realism are two competing criteria that must be reported. NowcastNet addressed intense precipitation by incorporating a mass conservation-motivated evolution operator into the generative network (Zhang et al., 2023).

2.3 Diffusion models, residual decompositions, and the latent problem

Denoising diffusion models (Ho et al., 2020; Song et al., 2021) learn to reverse a noising process and, in the formulation used here (Karras et al., 2022), are trained as a denoiser with an explicit pre-conditioning that ensures that the network’s effective regression target has unit-variance at each noise level. They are taught with a correct stable objective so they avoid the instability of adversarial training while still delivering sharp samples, and they naturally give an ensemble, as independent noise seeds give independent draws from the same conditional distribution.

Running diffusion at 256×256 is expensive, so latent diffusion is standard practice: compress the field with an autoencoder, diffuse in the latent space, and decode (Rombach et al., 2022). PreDiff applies this to nowcasting and, importantly for the codec designed here, trains its autoencoder with a pixel loss plus a patch adversarial loss and deliberately no perceptual loss, since no perceptual network pretrained on Earth observation data exists (Gao et al., 2023). LDCast does the same for radar with explicit attention to uncertainty quantification (Leinonen et al., 2023).

The notion of a residual arises in two ways. DiffCast decomposes a precipitation field into a global deterministic motion trend generated by a trained backbone and a stochastic residual handled by diffusion, and its ablations show that training the two concurrently is better than freezing the backbone (Yu et al., 2024). CorrDiff, designed for kilometre-scale downscaling rather than nowcasting, trains a regression network to predict the conditional mean, freezes it, then diffuses the remaining residual, using variance reduction as the stated reason (Mardani et al., 2025). This study takes the two-stage structure of CorrDiff, and examines what it adds when the base predictor is a physical scheme, not a learnt mean. Two more diffusion nowcasters broaden the horizon of what the family can achieve. Dai et al. (2025) extend a satellite-driven diffusion model to four hours, and Chen et al. (2025) employ a cascade of diffusion stages to keep convective cloud structure sharp at long lead. Both of them buy their horizon with way more computation than this project had. Neither of them distinguishes a non-learned physical component from the learnt one.

A cautionary outcome is that of Chase et al. (2025). They benchmarked eight off-the-shelf autoencoders on satellite infrared data, most of which they found to reconstruct badly. They tracked small-scale spectral noise in the latent diffusion output to the decoder, and suggested training a domain-specific autoencoder. They also tried a latent two-stage model of exactly the sort considered here, and gave it up without numbers. Both points are directly acted upon. A domain-specific codec is trained on radar (Section 4.2), its floor and ceiling are measured on the complete evaluation split, and the result that they left open is reported in Section 6.3.

2.4 Verification, and why a single score is misleading

The verification literature makes a point that dictates how everything is reported in this project. A pixel-wise score penalises a sharp forecast twice: a forecast of a shower 10 km away from where it fell is penalised once for being a miss and once for being a false alarm, but a blurred forecast poorly covering both locations is only penalised once (Roberts and Lean, 2008). This implies that mean absolute error and critical success index (Schaefer, 1990) consistently favour smooth fields: one can improve a model’s score by making it less realistic.

Three families of measure are used here to avoid that trap. The fractions skill score compares the fraction of exceedances inside a neighbourhood rather than pixel by pixel, and its crossing of a base-rate-dependent usefulness line gives the smallest scale at which a forecast is skilful (Roberts and Lean, 2008). The continuous ranked probability score, in its fair form for finite ensembles, rewards being close and being honest about uncertainty at the same time (Hersbach, 2000; Ferro, 2014). Unlike the pixel-wise measures above, it cannot be improved by purposefully leaning toward a smoother field because it is a proper score, which is optimised only by reporting the distribution the forecaster genuinely believes. Ensemble diagnostics, namely the ratio of spread to the root mean square error of the ensemble mean and the rank histogram, test whether the stated uncertainty is the right size (Fortin et al., 2014; Hamill, 2001). The spread ratio has a subtlety this project respects throughout: for an M -member ensemble the ideal is $\sqrt{M/(M+1)}$, not 1, which for $M = 8$ is 0.9428 (Fortin et al., 2014).

3 Data and problem setup

3.1 The radar product

The data is the Met Office UK radar rainfall composite, a national field merged from more than 15 radar stations (Ravuri et al., 2021) and distributed as ODIM HDF5 (Michelson et al., 2014) through a public AWS open-data bucket (Met Office, 2026). The grid is 2175×1725 pixels at 1 km spacing on a transverse Mercator projection over the British National Grid, with values as instantaneous rain rates in mm h^{-1} . Frames are sampled here on a 15-minute grid, which sets the forecast step everywhere downstream and makes +60 minutes exactly four extrapolation steps. The bucket holds a rolling window of roughly 19 months, so the ingest probes availability rather than assuming a date range. The record used here runs from 2024-11-21 to 2026-08-09.

The composite is Crown copyright and is redistributed openly through the AWS Open Data Registry (Met Office, 2026). It is an environmental observational product. It records rainfall rate over a national grid and contains no personal data, no human or animal subjects and nothing security-sensitive, so its use raises no data-protection question and required no ethical approval. It does have the attribution requirement which is satisfied by the above citation. The ethical questions this project does raise are not about the input but about the output, and they are taken up in Section 7.3.

The product uses three-valued logic that has to be respected before any statistic is computed. A raw value of -1.0 means **nodata**, the radar could not see that pixel, and is masked to not-a-number. A value of 0.0 means **undetected**, the radar looked and found nothing, and is genuinely dry. Anything positive is a measured rate. Treating **nodata** as dry would bias every mean, every wet-area fraction and every contingency count, so it is masked at read time and carried as an explicit validity mask through the whole pipeline. Rates are clipped at 128 mm h^{-1} , above which the composite is not trusted.

3.2 Crops, quality filter and the log transform

A full composite is far too large for a diffusion model, so the field is tiled into $256 \times 256 \text{ km}$ crops at 1 km resolution. Advection is computed on a 384×384 context window and then centre-cropped to 256, because rain blows into the scored box from outside and a 64-pixel margin covers a 15 m s^{-1} storm over an hour. A crop is kept only if at least 90 percent of its pixels are in radar range and at least 5 percent are wet at 0.1 mm h^{-1} . Of the 42 candidate tile positions on the national grid, 12 ever pass, and those 12 are shown in Figure 1. The filter excludes near-empty crops but does not reweight what is left, so the evaluation base rate is higher than a full-domain study would give, and comparisons

with published numbers have to be read with that in mind. How often each tile passes is given in Appendix A.

Rain rate is extremely skewed, mostly zero with rare large values, which destabilises both optical flow and diffusion. All modelling is therefore done in a logarithmic reflectivity-like domain, $\text{dB}(R) = 10 \log_{10} R$ for wet pixels, with dry pixels pinned to a floor of -15 dBR. The wet threshold is 0.1 mm h^{-1} . These three constants are fixed once and shared by every script in the pipeline, because changing one of them in one place would silently invalidate every comparison downstream.

3.3 Samples, splits and leakage

Each sample takes four consecutive past frames ending at a forecast origin t_0 , covering 45 minutes of history, and predicts the field at $+15$, $+30$, $+45$ and $+60$ minutes after t_0 . One lead-conditioned model covers all four leads.

The split is temporal and is judged on the date of the target frame rather than on the inputs, so no observation is ever a target in two splits at once. Validation is the first two available days of each calendar month, which samples every season rather than one contiguous block. Test is the whole of 2026, held out and scored once at the end, with the one departure from that rule recorded in Section 5. This is DGMR’s own scheme on the same product, doubled. DGMR assigns the first day of each month from 2016 to 2018 to validation and holds out 2019 entirely (Ravuri et al., 2021). Counts are in Table 1.

Each sample occupies a 105-minute window, 45 minutes of inputs before t_0 and a target 60 minutes after it, while the split is decided by the date of that target alone. A frame lying within 105 minutes of a split boundary can therefore enter one split as an input and another as a target. At a 15-minute step that is at most seven t_0 values per boundary, and with two boundaries per month over 21 months of record it bounds the exposure at roughly 1.3 percent of the 260,492 crops. In every such case the shared frame is an input to the training sample and never a training target, so the model is never asked to reproduce an observation it will later be scored on. A one-hour buffer either side of each boundary would close the path entirely and costs almost nothing, and it is the cheapest single improvement to the protocol.

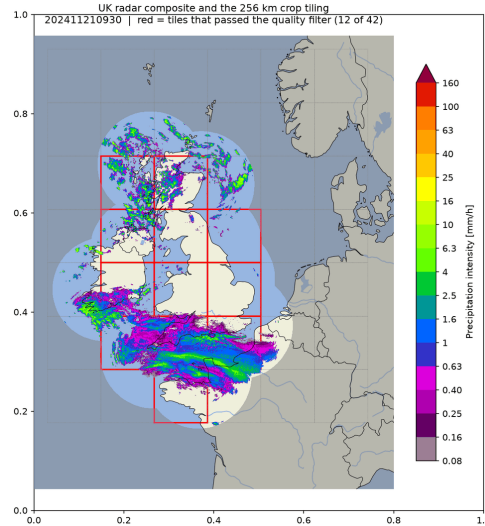


Figure 1: The evaluation domain: the Met Office composite grid with the 42 candidate 256×256 km tile positions, of which the 12 highlighted ever pass the quality filter. The other 30 sit mostly beyond radar range and would contribute empty crops.

3.4 The residual, and what there is to learn

Each cached crop holds the four inputs, the ground-truth target, the advection forecast, the residual $r = \text{dB}(y) - A_{\text{dBR}}$ and the validity mask. The ground truth is never discarded or replaced. Across 800 sampled crops the residual has a mean of $+0.18$ dBR and a standard deviation of 5.17 dBR. Those

Table 1: Dataset. Counts are quality-filtered 256×256 km crops. Each crop yields four lead-conditioned training samples, so the diffusion models see 613,892 training rows and 53,124 validation rows.

Split	Rule (judged on the target date)	Crops	Use
Train	everything not below	153,473	fitting all learned stages
Validation	first two available days of each month	13,281	selection and every comparison
Test	the 2026 calendar year	93,738	held out, used once, at +60
Total unique crops		260,492	1,041,968 cached samples, four leads

two numbers are the justification for the project stated compactly. A mean near zero says advection is an approximately unbiased first guess and therefore a sound anchor, and a spread of about 5 dBR says there is a large, structured signal left for a model to learn.

4 Method

The system has three stages: a non-learned advection prior, a frozen autoencoder that compresses rainfall fields, and a conditional diffusion model that learns the residual in the compressed space. Two arms share the first two stages and differ only in the third. Figure 2 shows both.

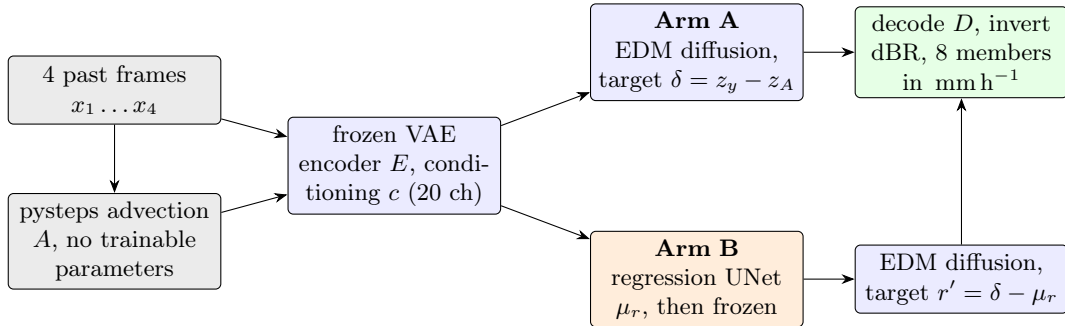


Figure 2: The pipeline. Grey is non-learned physics, orange is learned and then frozen, blue is trained, green is reconstruction. Both arms share the advection prior and the codec and differ only in what the diffusion model is asked to predict. Arm A assembles $\hat{z}_y = z_A + \delta$ and Arm B assembles $\hat{z}_y = z_A + \mu_r + \hat{r}'$. Eight independent noise seeds give an eight-member ensemble. Three outputs are reported throughout: the ensemble mean, one single member, and Arm B’s frozen regression mean $A + \mu_r$ on its own.

4.1 Stage one: the advection prior

The prior A is computed with pysteps (Pulkkinen et al., 2019), using its `dense_lucaskanade` motion estimator and its `semilagrangian` extrapolator. The dense local least-squares tracker is run on the four past frames in the dBR domain to give a velocity field, and the last frame is then transported forward by four 15-minute semi-Lagrangian steps, keeping the field at each of the four leads. If flow estimation fails, the velocity falls back to zero, which degrades the prior gracefully to persistence rather than producing nonsense.

Two properties of this stage are the point of the whole design. It has no trainable parameters, so it cannot overfit, cannot drift during training, and needs no validation budget of its own. And it is a deterministic function of the inputs alone, so it is computable at inference for any new sequence without reference to the training set.

4.2 Stage two: the codec

Diffusion at 256×256 was outside the compute budget, so a variational autoencoder (Kingma and Welling, 2014; Rombach et al., 2022) compresses each normalised dBR field to a $4 \times 64 \times 64$ latent, a fourfold spatial compression. The codec is trained on both y and A fields from every crop, because the diffusion stage has to encode both and a codec is only trustworthy on inputs drawn from its own training distribution. It is not trained on the residual, which is signed and therefore sits outside the distribution of non-negative rainfall fields. The residual is handled in latent space instead.

A first version trained with a masked L_1 loss and a tiny KL term reconstructed accurately on average and failed in exactly the way Chase et al. (2025) warn about. It retained only 45 percent of

observed power in the 2 to 8 km band, less than the raw advection field already carried, which would have capped the whole system below its own baseline at the scales of interest.

The second version fixes each failure with a change traceable to a named cause. Depth doubles to two residual blocks per resolution with self-attention at the bottleneck, following the PreDiff and Stable Diffusion encoder shape (Gao et al., 2023; Rombach et al., 2022). Transposed convolution is replaced by nearest-neighbour resize followed by convolution, which removes the checkerboard artefacts the first version showed. Those artefacts are the known consequence of a transposed convolution whose kernel size is not divisible by its stride, since the output pixels then receive unequal numbers of contributions, and resizing before convolving avoids the uneven overlap entirely. The pixel loss is weighted by rain intensity as $w = 1 + 23 \min(R, 24)/24$, which reproduces the clipped form of DGMR’s grid-cell weighting. DGMR prints $w(y) = \max(y+1, 24)$ but describes it in the same sentence as clipped at 24, so the function actually applied there rises linearly from 1 and saturates at 24, and the form used here is that one on a 0 to 24 mm h⁻¹ range (Ravuri et al., 2021). A hinge-loss patch discriminator is added, scoring overlapping receptive fields rather than the whole crop, and its contribution is scaled adaptively so that it can never overwhelm reconstruction. That is the codec recipe PreDiff and Stable Diffusion both use (Gao et al., 2023; Rombach et al., 2022), and the adaptive scaling is described in Appendix C. No perceptual loss is used, for the reason PreDiff gives, that no perception network pretrained on Earth observation data exists (Gao et al., 2023).

Epoch 17 of 25 is the selected codec, at validation weighted L_1 of 0.0331 and a 2 to 8 km power ratio of 0.957 against version one’s 0.454. Epoch 20 scored a higher power ratio and was deliberately not chosen, because it is the first epoch of adversarial collapse. Appendix C shows what the selected codec reconstructs, gives its measured bounds, and records the collapse diagnostic behind that decision.

4.3 Stage three, Arm A: direct residual diffusion

Everything is encoded once with the frozen encoder and cached. Writing s for the codec’s latent scale and $\mu(E(\cdot))$ for the encoder mean branch, each latent is $z = s \cdot \mu(E(\cdot))$, which puts the latents at approximately unit variance. Arm A’s target and conditioning are

$$\delta = z_y - z_A, \quad c = \text{concat}(z_{x_1}, z_{x_2}, z_{x_3}, z_{x_4}, z_A) \in \mathbb{R}^{20 \times 64 \times 64}. \quad (2)$$

The denoiser follows the EDM formulation (Karras et al., 2022), in which a raw network F_θ is wrapped in a preconditioning that makes its effective target unit-variance at every noise level:

$$D(x; \sigma, c) = c_{\text{skip}}(\sigma) x + c_{\text{out}}(\sigma) F_\theta(c_{\text{in}}(\sigma) x; c_{\text{noise}}(\sigma), c), \quad (3)$$

with $c_{\text{skip}} = \sigma_d^2/(\sigma^2 + \sigma_d^2)$, $c_{\text{out}} = \sigma\sigma_d/\sqrt{\sigma^2 + \sigma_d^2}$, $c_{\text{in}} = 1/\sqrt{\sigma^2 + \sigma_d^2}$ and $c_{\text{noise}} = \frac{1}{4} \ln \sigma$. Training draws $\sigma = \exp(P_{\text{mean}} + P_{\text{std}}n)$ with $n \sim \mathcal{N}(0, 1)$ and minimises $\lambda(\sigma)\|D(\delta + \sigma\epsilon; \sigma, c) - \delta\|^2$ with $\lambda(\sigma) = (\sigma^2 + \sigma_d^2)/(\sigma\sigma_d)^2$.

One choice here departs from common practice and matters. The constant σ_d has to be the standard deviation of the training target, and because that target is a residual rather than a field, its standard deviation is not 1. It was measured over the whole training pack as 0.7407. Defaulting to $\sigma_d = 1$, as many implementations do, would misweight the loss at every noise level, and CorrDiff’s released implementation hardcodes 0.5 for both of its stages rather than measuring either (Mardani et al., 2025). Measuring it is the reason validation losses are not numerically comparable between the two arms.

F_θ is a U-Net (Ronneberger et al., 2015) on 64×64 latents at resolutions $64 \rightarrow 32 \rightarrow 16$ with channel multipliers (1, 2, 4) on a base width of 128, two residual blocks per resolution, self-attention at 16×16 , and 65.7 M parameters. The noise level and the lead index enter through Fourier embeddings that drive a feature-wise affine modulation of every block’s group normalisation, predicting a per-channel scale and shift rather than being concatenated as extra input planes, so one model serves all four leads. Optimisation is AdamW (Loshchilov and Hutter, 2019) at learning rate 10^{-4} , 1000 warmup steps then cosine, batch 64, bf16, gradient clipping at 1.0, weight averaging at EMA decay 0.999, and

conditioning dropout of 0.1 to leave classifier-free guidance available (Ho and Salimans, 2022). Fifty epochs were run. Full settings are in Appendix B.

4.4 Stage three, Arm B: a frozen conditional mean underneath

Arm B inserts CorrDiff’s structure (Mardani et al., 2025) between the prior and the diffusion model. A deterministic network R_ϕ with the same architecture as F_θ , but with the noise input pinned, is trained to predict the conditional mean of the residual by plain unweighted mean squared error,

$$\mathcal{L}(\phi) = \mathbb{E} \|R_\phi(c, L) - \delta\|^2, \quad \text{whose minimiser is } \mu_r = \mathbb{E}[\delta \mid c, L]. \quad (4)$$

R_ϕ is then hard-frozen and never updated again, matching CorrDiff’s released implementation. The diffusion model’s target becomes the second residual $r' = \delta - \mu_r$, and μ_r is also appended to the conditioning stack, widening the U-Net input from 24 to 28 channels. This is CorrDiff’s `hr_mean_conditioning` setting. The paper’s text and its released configuration disagree on it, and this project follows the paper.

Because μ_r approximates a conditional mean, the law of total variance guarantees the second residual has no more variance than the first. The realised reduction was measured rather than assumed: explained variance on the training pack is 0.480, so σ_d falls from 0.7407 to $0.7407\sqrt{1 - 0.480} = 0.5341$, a 28 percent reduction in the spread of what the diffusion model must learn.

Before committing 26 GPU-hours to this arm, a 724-parameter ridge filter was fitted on 20,000 latent rows as a cheap lower bound on the predictable fraction, for about one CPU-hour. It reached a held-out explained variance near 0.33. Its feature decomposition is the more informative part. Using z_A alone gives 0.263 to 0.304, while using the past frames alone gives 0.011 to 0.025. Only the first of those two numbers supports an interpretation. It says that a linear map from the prior alone already explains about 30 percent of the prior’s own residual, which reads as a systematic shrinkage or decay correction rather than as new structure. The second number cannot be read as “the past frames carry nothing”, because A is itself a non-linear semi-Lagrangian warp of those same frames and a 3×3 linear filter cannot re-derive it, so a near-zero value there measures the filter’s limits and not the information content of the history.

Two properties of this design are deliberate. Everything except the target and the conditioning width is held identical to Arm A, so the comparison is a controlled single-change ablation. And the physical prior is retained in all three of its roles, as the additive anchor, as conditioning and as the reported baseline, so a weak μ_r degrades the arm back toward Arm A rather than toward an unanchored model.

4.5 Sampling and reconstruction

Sampling uses the deterministic second-order Heun sampler of Karras et al. (2022) with 25 steps, $\rho = 7$, $\sigma_{\min} = 0.002$, $\sigma_{\max} = 80$, guidance 1.0 and no stochastic churn. An eight-member ensemble is eight independent initial noise seeds through the same sampler. Each member is reconstructed by adding the sampled residual to the anchor in latent space, undoing the latent scale, decoding, undoing the codec normalisation and inverting the dBR transform:

$$\hat{y} = \text{dB}^{-1}(D((z_A + \mu_r + \hat{r}')/s) \cdot \text{std} + \text{mean}), \quad (5)$$

with μ_r omitted for Arm A. Three outputs are reported everywhere: the pixel-wise mean of the eight members, one single member, and for Arm B the frozen regression mean $A + \mu_r$ on its own, which is a complete deterministic forecast in its own right.

One assumption is exposed by this step and should be named. In pixel space $y = A + r$ is exact by definition, but the model works on $\delta = z_y - z_A$ and reconstructs through a non-linear decoder, so $D(z_A + \hat{\delta})$ recovers y only to the extent that the decoder is approximately linear over the range of the residual. That is an empirical assumption rather than an identity, and it is why the codec bounds are measured explicitly in Section 5.3.

4.6 Checkpoint selection

Choosing a diffusion checkpoint based on validation loss would be a mistake, and the reason is quantifiable. For all runs in this project, validation loss and small-scale power change in opposite directions at the end of training, thus the lowest-loss checkpoint is always the smoothest one generated by the run. For Arm A, the validation loss has remained within its own Monte Carlo noise floor since epoch 33 whereas over the same window the 2 to 8 km power ratio declined 5.4 percent and the heavy-rain critical success index fell 9.2 percent.

So a three-part rule was fixed in advance before either arm was scored: gate on sampled ensemble-mean error at or below the advection control on the same crops, then maximise the continuous ranked probability skill score, then break ties within 0.005 of that score on the member power ratio closest to the codec’s measured ceiling of 0.903. The rule applied independently to each arm’s ten archival checkpoints picks epoch 25 for both. Two arms with separate conditioning, different loss landscapes and different training histories converging at the same epoch is evidence that the rule picks on its criterion and not on noise. The choice is replicable from the training logs without a GPU and is listed in Appendix D. The two restrictions not disclosed by the outcome itself are given in Section 7.3.

5 Evaluation protocol

5.1 Baselines, pairing and what is scored against what

Each forecast is scored against y , the stored radar observation, at the pixels for which the validity mask indicates it is in range. The residual is always a target for training, never a target for evaluation, because to score $\hat{r}$ against r would vastly flatter the model by omitting the portion of the problem that physics already solves. Beside each model column are two baselines: persistence, the last input frame retained for the whole lead, and the pysteps advection prior. Both are calculated from the same crops using the same code thus are bit-identical across every scorecard in this report and every model-against-model comparison is matched on the same crops using the same masks.

Two provenance difficulties were encountered in scoring, as the two arms were scored at separate commits and one lead of the regression baseline was run later than the other three. Both were fixed by ensuring that the scorer and all modules that it imports are byte identical across the commits involved, rather than waiving the mismatch through. The check and the tool that enforces it are described in Appendix J.

5.2 Metrics

Five families are reported, chosen so that no single failure mode can hide.

Pixel error: Mean absolute error, root mean square error and bias, all in mm h^{-1} . Bias separates a genuinely accurate forecast from one that scores well by systematically under-forecasting.

Categorical skill: Critical success index, probability of detection and false alarm ratio at 0.5, 1, 2, 4 and 8 mm h^{-1} (Schaefer, 1990). Reporting per threshold is essential, because skill falls sharply with rain rate and the heavy end is the regime a nowcasting product exists for.

Spatial-scale skill: Fractions skill score at neighbourhood widths of 1, 5, 11, 21, 51 and 101 km (Roberts and Lean, 2008). The skilful scale is the smallest neighbourhood at which the score exceeds $0.5 + f_0/2$, with the base rate f_0 recovered from the contingency counts. Because the score is a ratio of sums, it is never averaged across leads or crops. Numerator and denominator are stored separately and pooled.

Ensemble quality: Fair continuous ranked probability score (Ferro, 2014; Hersbach, 2000), the ratio of ensemble spread to the root mean square error of the ensemble mean against its member-count-corrected ideal of $\sqrt{8/9} = 0.9428$ (Fortin et al., 2014), and the rank-histogram outlier rate against its ideal of $2/9 = 0.2222$ (Hamill, 2001).

Structure: Wet-area fraction at 0.1 mm h^{-1} , and the ratio of forecast to observed radially averaged spectral power in the 2 to 8 km band. Two estimators of that ratio disagree by roughly 7 percent on this data. The mean-of-ratios estimator is used throughout, and the two are never mixed inside one table.

5.3 The codec floor and ceiling

Because the model decodes through a frozen autoencoder, no result can be interpreted without knowing what that autoencoder alone costs and permits. Both bounds were measured on the full validation split rather than estimated, and Appendix C tabulates them. The ceiling is the observation itself put through an encode-decode round trip, at mean absolute error 0.0428 mm h^{-1} and a 2 to 8 km power ratio of 0.895 to 0.908 by lead, and no latent-space model here can exceed it in any principled sense. The floor is the advection field through the same round trip, 0.2672 mm h^{-1} against raw advection's 0.2656, so working in this latent space at all costs 0.6 percent of the baseline error before any learned component is added.

The ceiling has a consequence that is easy to get wrong and is applied throughout Section 6. Spectral power sitting above the ceiling cannot be recovered structure, because the ceiling is the observation itself through the same decoder, so any excess is synthesised high-frequency noise and counts as a sharpness defect rather than spectral skill. Power ratios are therefore read as distances from the measured ceiling, never as “closer to one is better”.

5.4 Threats to validity, and how each is handled

Small sample diagnostics, each arm logs an inexpensive 16-crop sampled diagnostic during training. In this project that statistic has now disagreed with full-split truth four times, twice in direction as well as magnitude, and it underestimates the ratio of member power by 15 to 32 percent. Hence it is viewed only as a tripwire for breakage. It is used nowhere else save for the checkpoint tiebreak of Section 7.3, and nothing in Section 6 depends on it.

Test set discipline: The rule was that the test split was scored once, at the end, on the frozen selection. This was broken in letter and not in substance, and the chronology is given rather than left for a reader to recreate. An early four-lead test evaluation of Arm A was run on 9 to 10 August 2026, at a checkpoint which neither selection procedure selects and before any selection has been made. The checkpoint rule was applied later, and only on validation evidence, thus no model was selected based on how well it looked on test, and that early scorecard is presented here as a sensitivity check only.

Codec confounding: An earlier single-lead model was trained on the first version of the codec and is not tabulated against any arm here, because the codec differs and the comparison would be uninterpretable.

6 Results

Pixel error, categorical skill and every probabilistic score below pool all 13,281 validation crops at each of four leads, 53,124 in total. Two families are cheaper to compute on a subsample and are, and the sample size travels with them everywhere it matters. The fractions skill score uses 403 crops per lead, taken as an even stride through the split, and the spectral power ratio uses 200 per lead, strided the same way and additionally restricted to crops that are more than 99 percent in radar range so that the spectrum is not contaminated by the mask edge. Those two carry RQ3 and the sharpness argument, so their smaller samples are stated again where the conclusion is drawn. Both diffusion arms are scored at `ckpt_ep025`, which is what the precommitted rule of Section 4.6 selects for each of them independently, with the sampler pinned identically at 8 members, 25 steps, guidance 1.0, no churn and seed 0. Pooled figures are given in the text below, and the per-lead breakdown behind them is in Appendix E.

6.1 The baseline, and what it leaves behind

Advection beats persistence comfortably on pooled error (0.2656 against 0.3396 mm h^{-1}) and on categorical skill at every threshold except the heaviest, where the two converge (0.0861 against 0.0708 at 8 mm h^{-1}). Its skill decays with lead as expected, from 0.1931 mm h^{-1} at +15 to 0.3231 at +60, with the critical success index at 1 mm h^{-1} falling from 0.559 to 0.321.

Figure 3 shows where the headroom is. In intensity, advection tracks the observed distribution up to roughly 5 to 10 mm h^{-1} and then falls below it, under-producing the heavy tail by a factor of two to three. In space it tracks the observed spectrum down to about 10 km and then departs, retaining 49

to 56 percent of observed power in the 2 to 8 km band. Persistence, which is a real radar frame simply held still, matches the observed histogram and spectrum almost exactly and is wrong only about where it puts the rain. That contrast is the cleanest available statement of what advection does, it moves rain to approximately the right place and smooths it on the way.

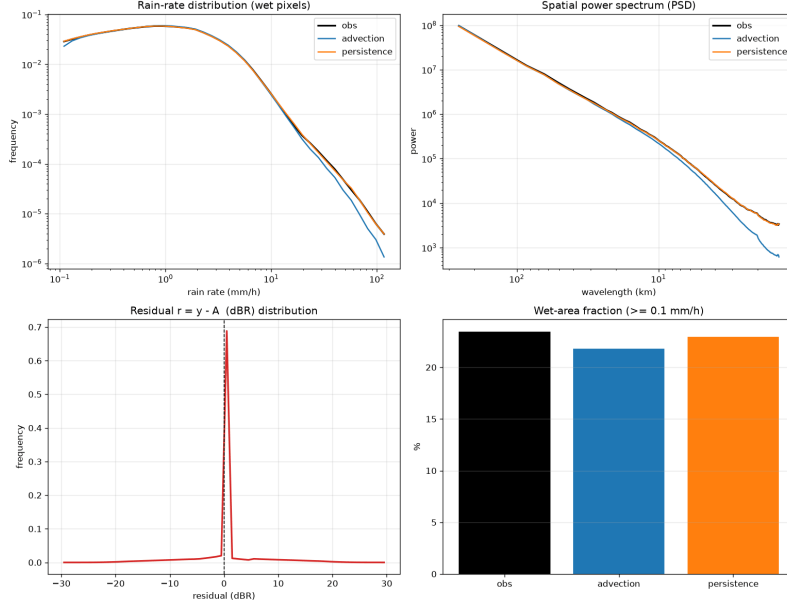


Figure 3: The advection baseline at +60 minutes on validation, 13,281 crops with the distributions taken from 2,000 of them. Clockwise from top left: rain-rate histogram over wet pixels, radially averaged power spectrum, wet-area fraction, and the residual r in dBR whose central spike is the large population of correctly forecast dry pixels. Persistence is a real radar frame, so it has the right texture and the wrong position, while advection has roughly the right position and loses both the heavy tail and the fine scales.

6.2 RQ1: does learning only the residual beat the physics?

Yes, and Table 2 is the answer. Every learned output beats advection on heavy-rain skill, and both ensemble means beat it on error as well. The headline is at the heavy end. The critical success index at 8 mm h^{-1} rises from advection’s 0.0861 to 0.1439 for the two-stage ensemble mean, a gain of 67 percent, and the regression mean reaches 0.1434 on the same crops. Fair continuous ranked probability scores of 0.1427 and 0.1386 against advection’s 0.2656 read as a point forecast give probabilistic skill scores of 46.3 and 47.8 percent.

Table 2: RQ1. Pooled validation scorecard, 53,124 crops, four leads, all methods on identical crops. PSD is the ratio of forecast to observed 2 to 8 km spectral power, as a range across the four leads, and it is the one column here computed on the 200-crop-per-lead spectral subsample rather than on all 53,124. Ens. says whether the method gives an ensemble. Observed wet-area fraction is 0.2305. Bias is identical between an arm’s mean and member rows, since bias is linear in the forecast.

Method	MAE ↓	RMSE ↓	Bias	CSI@1 ↑	CSI@8 ↑	Wet area	PSD	Ens.
Persistence	0.3396	1.2120	−0.0010	0.3002	0.0708	0.2292	n/a	no
Advection (pysteps)	0.2656	1.0177	−0.0164	0.4197	0.0861	0.2199	0.49–0.56	no
Regression mean (frozen stage)	0.1955	0.8099	−0.0796	0.4956	0.1434	0.1928	0.07–0.13	no
Arm A ensemble mean	0.2314	0.8191	+0.0056	0.4864	0.1329	0.3125	0.20–0.26	yes
Arm A single member	0.2848	1.0921	+0.0056	0.3992	0.0979	0.2278	0.92–1.02	yes
Arm B ensemble mean	0.2241	0.8067	+0.0064	0.5028	0.1439	0.3065	0.23–0.28	yes
Arm B single member	0.2757	1.0766	+0.0064	0.4175	0.1043	0.2277	1.00–1.07	yes
Codec oracle (ceiling)	0.0428	0.1940	+0.0071	0.8872	0.7378	0.2297	0.90–0.91	n/a

The table also shows that no output wins on every axis, and that is the most important qualitative result in the report. The regression mean is the best point forecast anywhere in the project, beating

even the better diffusion mean by 13 percent on error, and it is physically useless on its own: spectrally dead at 7 to 13 percent of observed fine-scale power, with its low error bought partly by systematic under-forecasting (dry bias $-0.0796 \text{ mm h}^{-1}$, wet area 0.1928 against an observed 0.2305). The diffusion ensemble means are close to unbiased but cover too much area (0.31 against 0.23) and remain smooth. Only the diffusion members are physically realistic fields, with wet area 0.2277 against 0.2305 and fine-scale power at or slightly above the codec ceiling. This is the double penalty of Section 2.4 made quantitative, and it is why every result here gives both the mean and a member. Appendix F shows the same split as distributions rather than as summary numbers.

The lead-time structure, in Figure 4, is the more interesting shape. Advection is nearly unbeatable at +15 and degrades faster than the learned forecasts do, so the learned component earns a growing share of the forecast the further ahead it looks. That growth is clean and monotonic for the regression mean, at 21.5, 25.4, 27.6 and 29.0 percent across the four leads, and for Arm B at 14.3, 15.5, 16.0 and 16.2 percent. Arm A does not follow it. Its margin peaks at +30 and narrows again to 12.6 percent, so the gap between the arms widens from 1.7 percentage points at +15 to 3.5 at +60. On heavy rain the effect is much larger, with Arm B more than doubling advection’s critical success index at +60, 0.0818 against 0.0398.

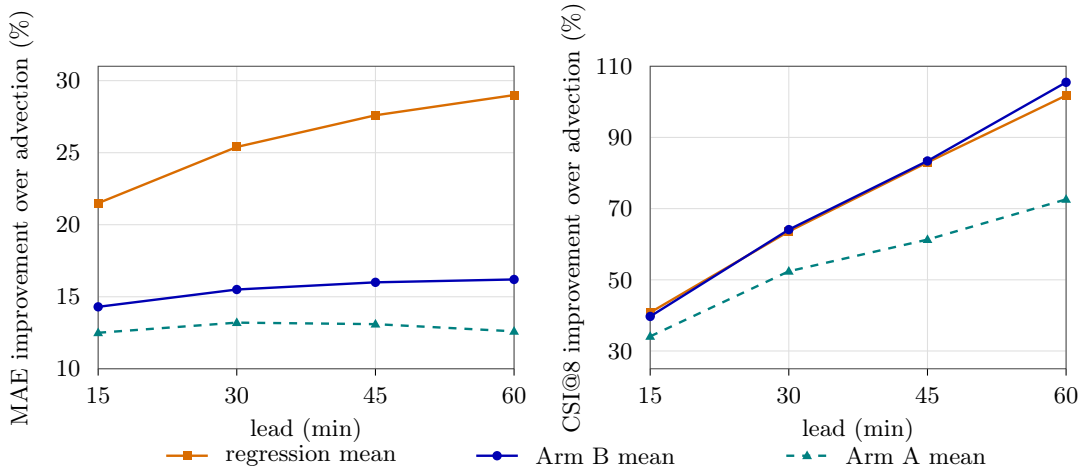


Figure 4: Skill over the advection baseline grows with lead time. Left: percentage reduction in mean absolute error relative to advection on the same crops. Right: percentage gain in the critical success index at 8 mm h^{-1} . Validation split, 13,281 crops per lead. Arm B’s margin widens monotonically, while Arm A’s peaks at +30 and then narrows again.

6.3 RQ2: does a frozen learned mean earn its place on top of the physics?

Yes on accuracy, and clearly not on calibration. Table 3 gives the comparison, and the ordering down the categorical rows is the result worth stating rather than any single number. Arm B is ahead on every accuracy column, and the margin rises monotonically with rain rate: 2.82 percent at 0.5 mm h^{-1} , 3.38 at 1, 4.71 at 2, 5.92 at 4 and 8.24 at 8, with no reversal in between. Whatever the frozen conditional-mean stage contributes, it is worth most where the field is rarest and hardest, which is the regime a nowcasting product is bought for. Arm B also wins at every individual lead on error, on both categorical columns and on the probabilistic score, so the pooled result is not an artefact of pooling.

Calibration is where the pre-registered expectation failed, and reporting that failure honestly matters more than the gain does. Before the arm was run, this project predicted in writing that the two-stage design would buy improvements in probabilistic score, error and calibration. The first two are confirmed. The third is not: at epoch 25 both arms sit within a quarter of a percent of the ideal spread ratio, 0.9426 and 0.9449 against 0.9428, and their outlier rates differ by 0.003. That difference is smaller than the spread across an arm’s own individual leads, so nothing in this evidence supports ranking the two arms on dispersion.

An earlier version of this analysis did claim a calibration gain, that the two-stage arm cut the dispersion deficit from 3.1 percent to 0.6. That claim is withdrawn. It was an artefact of an asymmetric

Table 3: RQ2. Arm A against Arm B, both at the rule-selected epoch 25, pooled over 53,124 validation crops on identical crops. The ideal spread ratio for an 8-member ensemble is $\sqrt{8/9} = 0.9428$ and the ideal outlier rate is $2/9 = 0.2222$.

Pooled column	Arm A (direct)	Arm B (two-stage)	Arm B better by
Ensemble-mean MAE ↓	0.2314	0.2241	3.15 %
Ensemble-mean RMSE ↓	0.8191	0.8067	1.52 %
CSI @ 0.5 mm h ⁻¹ ↑	0.5509	0.5664	2.82 %
CSI @ 1 mm h ⁻¹ ↑	0.4864	0.5028	3.38 %
CSI @ 2 mm h ⁻¹ ↑	0.3719	0.3894	4.71 %
CSI @ 4 mm h ⁻¹ ↑	0.2422	0.2565	5.92 %
CSI @ 8 mm h ⁻¹ ↑	0.1329	0.1439	8.24 %
Single-member MAE ↓	0.2848	0.2757	3.19 %
Single-member CSI @ 8 mm h ⁻¹ ↑	0.0979	0.1043	6.52 %
Fair CRPS ↓	0.1427	0.1386	2.85 %
Spread / RMSE (ideal 0.9428)	0.9426	0.9449	a dead heat
Outlier rate (ideal 0.2222)	0.2600	0.2630	a dead heat

checkpoint pairing, comparing Arm B at epoch 15 against Arm A at its over-smoothed last archived epoch 50, so the deficit being cut belonged to that one checkpoint rather than to the architecture. Applying the same precommitted rule to both arms removed it, and the correction moved the headline in Arm B’s favour on everything else, with the heavy-rain margin rising from 2.5 to 8.24 percent. The old pairing understated the arm on accuracy while overstating it on calibration.

The latent measurement behind all this is the number Chase et al. (2025) left unreported. The frozen regression stage reaches a held-out explained variance of 0.461, rising monotonically with lead from 0.425 at +15 to 0.477 at +60, and 0.480 on the training pack. So roughly half the variance of the latent residual is deterministically predictable from the prior and the recent past, and the predictable fraction grows with lead. Two caveats travel with that figure. It is a latent-space variance ratio, and because the decoder is non-linear it bounds nothing directly in mm h⁻¹, so its pixel-space consequence is the decoded regression-mean row of Table 2, which has to be quoted alongside it. The network also overfits from epoch 3 onward, with training explained variance rising to 0.570 while validation falls to 0.427, so the reported checkpoint is epoch 2 of 8.

6.4 RQ3: what does averaging the ensemble cost?

A great deal, and the cost is spatial resolution. Figure 5 gives the fractions skill score against neighbourhood width, and Table 4 reduces it to the number a forecaster would actually ask for, which is the smallest neighbourhood at which each output is skilful. Both rest on the 403-crop-per-lead fractions-skill subsample rather than on the full split, so this is the thinnest evidence base behind any headline in the report, and the margins here should be read accordingly.

Table 4: RQ3. Smallest skilful neighbourhood width in km, that is, the smallest scale at which the fractions skill score exceeds $0.5 + f_0/2$. Arm B, validation split, 403 crops per lead taken as an even stride. Lower is better. Available widths are 1, 5, 11, 21, 51 and 101 km. The target is 0.547 at 1 mm h⁻¹ and 0.501 at 8 mm h⁻¹, from base rates of 0.094 and 0.0022 recovered from the contingency counts on the full split.

Threshold	Output	+15	+30	+45	+60
1 mm h ⁻¹	Persistence	1	5	11	11
	Advection	1	1	1	5
	Ensemble mean	1	1	1	1
	Single member	1	1	1	5
8 mm h ⁻¹	Persistence	21	51	51	101
	Advection	5	11	21	51
	Ensemble mean	5	11	21	51
	Single member	5	11	21	21

At light rain there is very little tension. The ensemble mean resolves 1 mm h⁻¹ rain at the grid scale at every lead, and a single member does the same out to +45 before it needs a 5 km box at +60, which

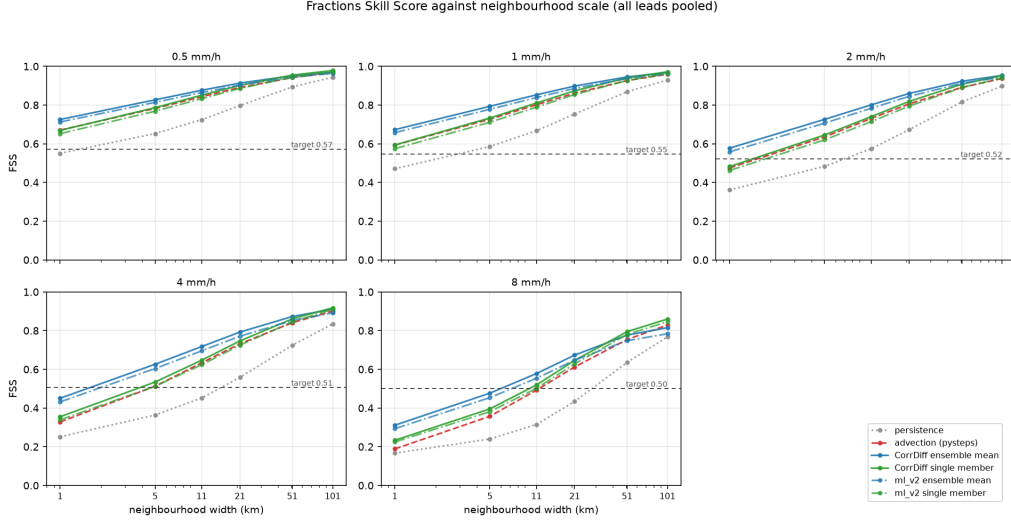


Figure 5: RQ3. Fractions skill score against neighbourhood width, pooled over all four leads, one panel per threshold. The dashed line is the usefulness target $0.5 + f_0/2$ for that threshold’s observed base rate, which runs from 0.572 at 0.5 mm h^{-1} down to 0.501 at 8 mm h^{-1} . At light rain the ensemble mean dominates at every scale. At 8 mm h^{-1} it leads only out to 21 km and the single member overtakes it somewhere between 21 and 51 km, and that crossing is RQ3 in one picture. The full pooled table is in Appendix H.

is the box advection needs there too. Persistence needs 11 km from +45 onward. The tension appears at heavy rain and the longest lead, which is where a nowcast is most needed and hardest to make. There, persistence needs a 101 km box before it is useful and advection needs 51. A single ensemble member needs 21 km, roughly two and a half times finer than the physics baseline. The ensemble mean of the very same members needs 51 km, identical to advection. Averaging throws away precisely the resolution the model had gained.

This is not a defect of the ensemble mean, which is the better forecast on every pointwise score in Table 2. It is a statement that the two outputs answer different questions and that reporting either alone misrepresents the system: the mean is the right product when a single best-estimate field is wanted and the user will aggregate anyway, and a member is the right product when the spatial structure is what the decision depends on.

6.5 Holding on an unseen year

The validation split is temporally interleaved with training, so it tests generalisation across days but not across years. The 2026 test split is a genuine temporal holdout: both arms were trained on data ending 2025-12-31, and the test period runs 2026-01-01 to 2026-08-09. Table 5 reports it at +60, with 13,281 crops taken as an even stride through the 93,738 available so the score is not weighted toward the earliest months.

Table 5: Held-out 2026 test split at +60 minutes, both arms at the rule-selected epoch 25, 13,281 crops stride-sampled from 93,738. Observed wet-area fraction 0.2261.

Method	MAE ↓	RMSE ↓	CSI@1 ↑	CSI@8 ↑	Fair CRPS ↓	Spread/RMSE	Wet area
Persistence	0.3369	1.3272	0.2076	0.0353	n/a	n/a	0.2231
Advection (pysteps)	0.2785	1.1829	0.2963	0.0449	n/a	n/a	0.2081
Arm A mean	0.2479	0.9478	0.3611	0.0559	0.1522	0.8921	0.3338
Arm B mean	0.2384	0.9314	0.3817	0.0659	0.1472	0.8928	0.3255
Arm A member	0.2986	1.2345	0.2774	0.0467	n/a	n/a	0.2198
Arm B member	0.2870	1.2134	0.2977	0.0526	n/a	n/a	0.2209

RQ1 holds out of sample. Arm B beats advection by 14.4 percent on error, 28.8 percent on the critical success index at 1 mm h^{-1} and 46.5 percent at 8 mm h^{-1} , and Arm A by 11.0, 21.9 and 24.4 percent on the same three. RQ2’s answer holds and strengthens. Arm B leads Arm A by 3.83 percent on ensemble-mean error against 3.15 on validation, by 5.70 percent at 1 mm h^{-1} against 3.38, by 17.77

percent at 8 mm h^{-1} against 8.24, and by 3.26 percent on the probabilistic score against 2.85. Every one of those margins is larger out of sample than in, and the growth is largest at heavy rain. A margin that widens on a year the model has never seen is much harder to explain as overfitting than one that shrinks, and this is the strongest framing the evidence supports.

Two caveats travel with the table and are not buried. Both arms are about 5.3 percent under-dispersed on test, at 0.8928 and 0.8921 against the 0.9428 ideal, where at the same lead on validation they were 0.5 and 0.8 percent from ideal. Parity between the arms survives but calibration itself does not, and the deficiency is shared rather than arm-specific. The ensemble does not become narrower out of sample, since spread is essentially unchanged between the two splits, falling only 0.3 percent for Arm B and 0.1 for Arm A. What moves is the error, with the ensemble-mean root mean square error rising 4.8 and 4.7 percent. The spread simply fails to grow with it. That is consistent with 2026 having a heavier tail than the validation period while being easier on average, which the baselines show directly at the same lead, since advection’s mean absolute error falls from 0.3231 to 0.2785 mm h^{-1} while its root mean square error rises from 1.1404 to 1.1829. Appendix F shows the same thing from the other side. The second caveat is that this pairing exists at +60 only, so every lead-time claim in this report is a validation claim.

A checkpoint-sensitivity check sits underneath the table and is reported because it is inconvenient. The early four-lead test evaluation described in Section 5 scored Arm A at epoch 35, where it reaches 0.2448 mm h^{-1} at +60 against 0.2479 at the rule-selected epoch 25. Epoch 35 is therefore about 1.3 percent the better Arm A checkpoint on held-out data even though the rule, run on validation, picks epoch 25. The selection rule is not optimal in hindsight, which is exactly what an out-of-sample check is for. The arm ranking is unaffected: Arm B at epoch 25 still beats Arm A at its best measured test checkpoint by 2.6 percent on error and 10.6 percent at 8 mm h^{-1} .

6.6 Case studies: the mechanism, and where it reverses

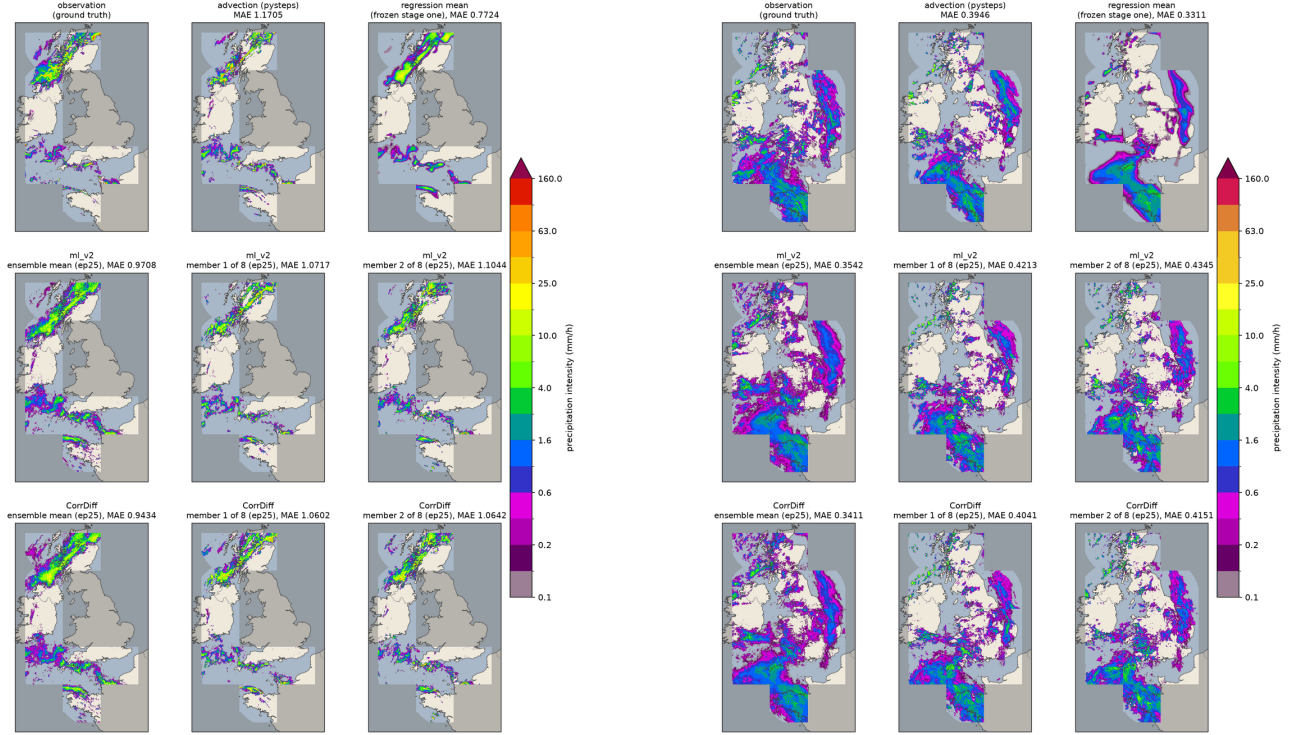
Six frames from the held-out test year show the mechanism rather than assert it. Two are given in full detail in Figure 6 and Table 6, all six are summarised in Table 7, and the remaining four are plotted at the same size and with the same panel layout in Appendix G. Each frame is a mosaic of the accepted tiles at one timestamp, so they are geographically real but they are not full-composite nowcasts. The six were selected to span both regimes and the range of rain amount present in the test year rather than sampled at random, which bounds what they can be asked to support.

Table 6: The two case studies. MAE in mm h^{-1} , lower is better. The “vs adv” column is the percentage improvement on the advection baseline for that same frame, and wet area is the fraction of valid pixels at or above 0.1 mm h^{-1} .

Output	Convective, 25 Jun 2026 22:30				Frontal, 15 Feb 2026 05:30			
	MAE	vs adv	Wet area	vs obs	MAE	vs adv	Wet area	vs obs
Regression mean	0.7724	+34.0%	0.1817	−10.8%	0.3311	+16.1%	0.3099	−30.0%
Arm B mean	0.9434	+19.4%	0.3265	+60.3%	0.3411	+13.6%	0.5156	+16.4%
Arm A mean	0.9708	+17.1%	0.3149	+54.6%	0.3542	+10.2%	0.5433	+22.6%
Arm B member 1	1.0602	+9.4%	0.2001	−1.7%	0.4041	−2.4%	0.3829	−13.6%
Arm A member 1	1.0717	+8.4%	0.1863	−8.5%	0.4213	−6.7%	0.3883	−12.3%
Advection	1.1705	baseline	0.1740	−14.6%	0.3946	baseline	0.3705	−16.4%
Persistence	1.4956	−27.8%	0.2203	+8.2%	0.4633	−17.4%	0.3987	−10.0%
Observed	n/a	n/a	0.2037	n/a	n/a	n/a	0.4430	n/a

The convective night is the regime advection is built to fail in, which is the cells that grow and decay rather than simply translate. Every learned output beats it, the ensemble mean by 19.4 percent and a single member by 9.4, and the member’s wet-area fraction lands within 1.7 percent of observed where advection is 14.6 percent too dry. The frontal morning is the opposite, widespread rain that mostly translates, which is what Lagrangian extrapolation does well. The ensemble means still win by 13.6 and 10.2 percent, but every individual member of both arms loses to advection, by 2.4 to 10.1 percent.

Read across all six frames the regime reading does not survive, and the four additional frames



(a) Convective, 25 June 2026 22:30 UTC, 8 tiles.

(b) Frontal, 15 February 2026 05:30 UTC, 12 tiles.

Figure 6: Two held-out test frames at +60 minutes. Top row: observation, advection, frozen regression mean. Middle row: Arm A ensemble mean and two members. Bottom row: Arm B ensemble mean and two members. In the convective case advection has smeared the cells and weakened them and every learned output improves on it. In the frontal case advection is already close to right, the ensemble means still win, and the individual members lose.

Table 7: All six case frames, ordered by how much error the advection prior leaves behind. C is convective and F frontal, B and A are the two arms, and mem. is a single member. Columns 3 to 6 give the percentage improvement in MAE on that frame’s own advection baseline, so positive beats the physics. The last two columns give wet-area fraction as a percentage difference from observed, so positive is too wet. The frames marked † are the two in Figure 6, and the other four are in Appendix G.

Frame (2026)	Adv MAE	MAE, % better than advection				Wet area, % vs obs.	
		Reg. mean	B mean	A mean	B mem.	B mean	B mem.
C, 19 Jun 18:30	0.1124	+29.3	+1.0	−1.5	−0.8	+63.5	−7.6
F, 15 Feb 05:30†	0.3946	+16.1	+13.6	+10.2	−2.4	+16.4	−13.6
C, 2 Jun 15:45	0.4870	+32.1	+6.4	+8.0	+4.6	+65.3	+2.1
F, 11 Jan 19:15	0.5827	+22.2	+22.7	+20.8	+7.0	+29.3	−3.7
F, 27 Jan 05:00	0.7414	+23.5	+24.8	+21.2	+10.6	+14.1	−11.1
C, 25 Jun 22:30†	1.1705	+34.0	+19.4	+17.1	+9.4	+60.3	−1.7

are what overturn it. Ordered by how much error advection leaves behind, the gain from the Arm B ensemble mean rises from 1.0 percent on the lightest frame to between 19 and 25 percent on the three heaviest (Spearman $\rho = 0.77$), and it tracks the observed wet fraction more tightly still ($\rho = 0.94$). Split by regime instead, the frontal frames average a 20.4 percent gain against 8.9 for the convective ones, which is the opposite ordering to the one two frames implied. Regime was confounded with rain amount, because the three frontal frames carry 44 to 58 percent wet area and the three convective ones 10 to 20 percent. What looked like a property of convection reads better as a property of how much there was to fix.

The 19 June frame is the boundary condition of the whole design. Advection is already accurate there to 0.11 mm h^{-1} , so Arm B gains only 1.0 percent, Arm A loses 1.5, and the two ensemble means put down 63 and 69 percent more wet area than was observed. When the physics is already right the diffusion has no error to correct and contributes only drizzle. Section 2.1 warned that any comparison against advection risks averaging over regimes where the baseline is unbeatable and regimes where it is weak, and this is that first regime caught at frame level rather than assumed. The pooled scores average it away entirely.

Three things hold on every frame. Arm B’s ensemble mean beats advection in all six and beats its own single member in all six, by 1.8 to 16.0 points. Every ensemble mean is too wet, by 14 to 69 percent, while every single member sits within 13.6 percent of the observed wet fraction, so for Arm B the mean absolute wet-area error is 41.5 percent for the mean against 6.6 for its members. That is the RQ3 trade in one line, now shown on six frames rather than argued from two. Arm B also beats Arm A on five of the six, 2 June being the single reversal, which is worth stating because the pooled tables report that advantage without exception. Six hand-picked frames are still not statistics, and the rank correlations rest on six points chosen partly for their spread, so they are offered as a mechanism worth testing rather than as a measured relationship.

6.7 Hyperparameter search

Three studies screened 43 distinct configurations across nine parameters as 56 trial-runs for about 22 GPU-hours, the two training screens ladderred by successive halving with $\eta = 3$ (Li et al., 2018) against a naive full-fidelity equivalent of 967 A100-hours. The verdict is that the incumbent configuration was already near-optimal, and nothing found is worth retraining, which changes the defaults used into the defaults that survived a screen. Width 192 is the only architectural change that helps, and it was not adopted because it would mean retraining both arms and everything downstream of them. Classifier-free guidance (Ho and Salimans, 2022) is a clean negative result on the highest-leverage untried knob, degrading the probabilistic score monotonically while dispersion collapses from 0.916 at guidance 1.0 to 0.818 at 1.5. Regularisation does nothing measurable in either screen despite the regression network overfitting from epoch 3. Full effects are in Appendix I.

One methodological finding constrains how the search may be reported, and it was measured rather than assumed. The ladder is based on rankings that transfer between fidelity rungs. The Spearman rank correlation between the two lowest rungs is +0.38 on 8 trials for the U-Net screen and -0.50 on 3 trials for the regression screen. Neither supports the assumption, so the ranking the search produces is not quotable, although the marginal effects still are, because each of those is a paired single-variable change at a fixed fidelity.

One qualification belongs alongside the no-retrain decision. The U-Net ladder pushed only two configurations to the deepest rung, both from the learning-rate-by-batch block rather than the one-factor arm. Batch 32 is the only setting in the screen with deep-rung evidence, and neither arm was trained on it. The deepest rung is still a six-epoch screen and cannot be compared to a production scorecard. Therefore, this is not evidence against the incumbent. The screen cannot fully resolve the incumbent, and batch size remains the only viable candidate after the search.

7 Discussion

7.1 What the results mean for the research questions

RQ1 is answered favourably and out of sample. A deterministic component with no free parameters is a sufficient basis for a residual diffusion nowcaster. On a year the models never saw, the two-stage arm improves on the physics it is built from by 14.4 percent on error and 46.5 percent on heavy-rain skill, and the direct arm by 11.0 and 24.4 percent. The claim that should be made from this is narrow and defensible, which is that freezing a parameter-free physical prior does not cost what freezing a learned backbone costs. It is not a claim that a physical prior beats a learned one, because this project never trained a learned deterministic backbone in place of A and therefore cannot make that comparison.

RQ2 is answered favourably but narrowly, and the shape of the answer is more informative than its sign. A frozen learned mean does earn its place on top of a parameter-free prior, but it buys accuracy rather than calibration and it buys most of that at heavy rain. The most plausible mechanism is suggested by the ridge-filter decomposition of Section 4.4: a linear map from the prior alone already recovers about 30 percent of the prior’s own residual, which reads as a shrinkage and decay correction applied to the advection field. That is evidence about what is linearly available in z_A , not a demonstration that the trained network confines itself to such a correction, and the companion x-only number cannot be added to the argument for the reason given there. Advection’s characteristic error at heavy rain is exactly over-persistence of intensity in decaying cells and under-production in growing ones, so a correction of that form should help most where intensity is extreme, which is what the monotone ordering across the categorical rows shows.

RQ3 is answered with a number that differs depending on which table it is read from. Collapsing an eight-member ensemble to its mean improves every pointwise score and moves the smallest skilful scale under heavy rain from 21 km back to the 51 km of the physics baseline. The skill evident in the pointwise table and the skill visible in the scale table are not the same. Reporting simply the first would describe a system that is no better resolved than extrapolation at the scales relevant to convective forecasting.

7.2 Relation to the literature

The available comparisons are directed rather than like-for-like, and they are described accordingly. DGMR trained on the same Met Office 1 km composite in 256 km crops, held out a full year, and reports significant gains over pysteps on the critical success index at every threshold (Ravuri et al., 2021). The direction of result here agrees, on the same product and a similarly structured holdout, from a very much smaller compute budget. DGMR also reports effective resolutions of 32 km for its U-Net baseline and 16 km for an axial-attention baseline at +90 minutes, which brackets the 21 km single-member figure found here at +60 and suggests the scale result is in a plausible range rather than anomalous.

The relationship to DiffCast is the sharpest one. DiffCast froze a learned deterministic backbone and lost skill relative to training it jointly, dropping the critical success index from 0.3077 to 0.2739 (Yu et al., 2024). This project freezes a deterministic component that has nothing to learn, and the residual arm built on it beats that component substantially. The two results are not in conflict. DiffCast measured the cost of preventing a trainable component from adapting to its partner, not the cost of the two-stage structure. That is provided as a hypothesis supported by two data points, not a demonstration.

CorrDiff keeps its regression mean frozen and beats that stage on probabilistic score (Mardani et al., 2025). The two-stage arm here also beats its own frozen stage on probabilistic score and on realism, while losing to it on pointwise error. CorrDiff reports ensemble under-dispersion as its own honest caveat, and the same deficiency appears here out of sample, which suggests it is a property of the recipe rather than of this implementation. Finally, Chase et al. (2025) abandoned a latent two-stage model without numbers, and this project supplies them. Their codec warning was well founded, though. The first codec trained here failed exactly as they describe, and only the adversarially trained replacement made the pipeline viable.

7.3 Limitations

The held-out evidence is single-lead. The reported test pairing exists at +60 only. The lead-time trend of Figure 4 is a validation result and is quoted as one. No held-out lead-time trend exists for the two-stage arm at all.

The checkpoint rule has two weaknesses the outcome hides. Its tiebreak consumed the 16-crop in-training diagnostic, which Section 5 shows understates the member power ratio by 15 to 32 percent, so the rule chose on a biased input. The two arms are also not equally robust to the tie tolerance. For Arm A the nearest excluded checkpoint would not change the outcome. For Arm B, a tolerance of 0.0064 rather than the precommitted 0.005 would have selected epoch 15, whose distance from the power ceiling is 3.1 times smaller. The defence is that 0.005 was fixed before either arm was scored rather than chosen afterwards, but that is a weaker claim than robustness.

Both arms over-sharpen at the reported checkpoint. Member power sits at 103 to 119 percent of the measured codec ceiling depending on arm and lead. Since the ceiling is the observation itself through the same decoder, the excess is synthesised high-frequency noise rather than recovered structure, and the two-stage arm produces more of it. The closest any checkpoint measured in this project comes to the ceiling is Arm A at epoch 50, which sits below it at the two short leads and 1 and 5 percent above it at +45 and +60, and which is also the checkpoint whose ensemble is 3.1 percent under-dispersed. Sharpness and dispersion trade against each other along the epoch axis, and no checkpoint in this project gets both right.

The explained-variance figure is a latent quantity. Because the decoder is non-linear, $D(\mathbb{E}[z]) \neq \mathbb{E}[D(z)]$, so the 0.461 explained variance bounds nothing directly in mm h^{-1} . Its pixel-space consequence is measured separately and both are reported, but the cleanest experiment separating “the mean is unpredictable” from “the mean is predictable and the decoder destroys it” would be a pixel-space regression at 256×256 , which the compute budget did not allow. Relatedly, the diagnostic variant that predicts the target latent directly rather than the residual differs from the reported variant in both its target and its conditioning mode, so although the residual formulation wins at every lead by 1.8 to 2.4 percent, the two effects cannot be separated.

The two scale-sensitive metric families are computed on subsamples. Pixel error, categorical skill and the probabilistic scores use all 53,124 validation crops, but the fractions skill score uses 403 crops per lead and the spectral power ratio 200, both strided evenly through the split. That is a deliberate cost decision rather than an oversight, and the strided construction keeps them representative in time, but it means the RQ3 headline and every power ratio quoted here rest on roughly three percent and one and a half percent of the split respectively. Recomputing the fractions skill score on the full split needs no GPU, since the samples are stored, and it should be done before the 21 km against 51 km result is relied on outside this report.

No margin in this report carries an uncertainty estimate. Every comparison above is a point estimate. The crops are also not independent, the 13,281 validation crops at each lead are twelve simultaneous tiles of the same weather sampled every 15 minutes, so the effective sample size is well below the nominal 53,124 and an interval computed as though the crops were independent would be too narrow. This matters most for the smallest margin, the 2.82 percent at 0.5 mm h^{-1} , on which the monotone ordering in Section 6.3 leans hardest. The 8.24 percent at 8 mm h^{-1} is large enough that noise is unlikely to reverse it.

Scope: The proposal promised a DGMR comparison and a storm-speed stratification over a three-hour horizon, and neither survived the compute budget. The horizon is +15 to +60 minutes against pysteps and persistence, and the DGMR comparison is directional, drawn from published figures rather than from a retrained model. The crop filter also means the evaluation base rate is higher than a full-domain study would give, so absolute scores here are not directly comparable to full-composite numbers in the literature.

The ethical exposure sits in the output, not the input. The two described products fail in opposing directions, with implications for road, aviation, and flood considerations. The ensemble mean covers 0.31 of the area compared to the observed 0.23, resulting in an over-warning forecast that users tend

to ignore. The observed wet proportion is matched by a single member, but the rank histogram of Appendix F shows a quarter of wet pixels above all eight members. This suggests that the models place too little rain, with the deficit in the upper tail. The reporting rule outlined in Section 6.4 is not just scientific but also ethical, as a 51 km skilful scale read as a 1 km field is the most likely to cause injury. The pipeline is not forecasted and is based on historical radar data.

7.4 Future work

Four items follow directly from the limitations, in the order their cost and value suggest. Extend the held-out evaluation to all four lead times, which needs no new training and would turn the lead-time trend from a validation claim into a held-out one. Act on the dispersion diagnosis of Section 6.5: because the spread does not respond when the error grows, the remedy is a mechanism that lets it, such as stochastic churn in the sampler, which the search showed reaches near-ideal dispersion but overshoots the spectral ceiling, or an explicit post-hoc spread calibration fitted on validation. Confirm the width-192 result at full fidelity, which the search identifies as the only architectural change worth having and which the multi-fidelity transfer failure means cannot be claimed from the screen alone. And beyond this project’s frame, the DiffCast hypothesis of Section 7 suggests the experiment that would settle it: train the conditional-mean stage jointly with the diffusion model on top of the same parameter-free prior, and measure whether unfreezing a learned mean helps here as it did for DiffCast, or whether the physical anchor makes joint training unnecessary.

8 Conclusion

This project asked whether a diffusion nowcaster can be built on a deterministic component with no parameters at all, and whether a learned mean is still worth stacking on top of one. Both answers are yes, with the qualifications set out above.

A parameter-free pysteps advection prior is a sound anchor for a residual latent diffusion model. On a held-out 2026 year the two-stage arm improves on that prior by 14.4 percent on mean absolute error and 46.5 percent on the critical success index at 8 mm h^{-1} , and the direct arm by 11.0 and 24.4 percent, with fair probabilistic skill scores near 47 percent against the prior read as a point forecast. A frozen learned conditional-mean stage added on top earns its place, it wins on every accuracy column by a margin that rises monotonically with rain rate, from 2.8 percent at 0.5 mm h^{-1} to 8.2 at 8 mm h^{-1} on validation and to 17.8 percent out of sample. What it does not buy is calibration, which was half of the written prediction made before the arm ran and did not survive symmetric checkpoint selection. And collapsing the ensemble to its mean costs real spatial resolution, moving the smallest skilful scale at heavy rain from 21 km back to the 51 km of the physics baseline, so both outputs must be reported.

The contribution, stated at the width the evidence supports. DiffCast found that freezing a learned deterministic backbone hurt. A deterministic backbone with nothing to learn does not hurt, and a learned mean on top of it still adds value, which is the case neither DiffCast nor CorrDiff tested.

Data, code and artefact availability

Everything this report claims can be recomputed is released. The archive holds the stage scripts of Appendix J, the selected codec and both diffusion checkpoints, every scorecard JSON behind Tables 2 to 7, the per-crop fractions-skill numerators and denominators, the six case-study JSON files, the hyperparameter trial log, and the plotting scripts that regenerate every figure here from those files with no GPU. The radar itself is not redistributed, because it is Crown copyright and already public, and the ingest script rebuilds the crop cache from the AWS Open Data bucket given a date range (Met Office, 2026).

The archive is held on University of Exeter OneDrive at https://universityofexeteruk-my.sharepoint.com/:f:/g/personal/dv321_exeter_ac_uk/IgB4eT3B00PgQaA71_4yJismAdfNnq8lQhOuTXJSt0hVKyo?e=PdxQ1s.

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Appendices

The ten appendices below hold the material the 20-page main text cannot carry. None of them is needed to follow the argument, and each is referred to from the point in the main text that depends on it. Appendices A and B shows the preprocessing contract every script shares, and the full configuration of every learned stage. Appendix C bounds what the frozen codec permits, which is the constraint every result in Section 6 is read against. Appendix D reproduces the checkpoint decision from the training logs, so a reader can rerun the selection rule by hand. Appendices E to H are the evidence behind the four headline results: the per-lead scorecards, the ensemble calibration diagnostics, the four case frames not printed in Section 6.6, and the full fractions-skill table. Appendix I gives the hyperparameter search in full, and Appendix J says how to rebuild all of it.

A Data, preprocessing and the crop filter

Every constant below is fixed once in the pipeline and shared by every script that touches the cache. Changing any one of them in one place, and not in the others, would silently invalidate every comparison downstream, so they are recorded here as the contract rather than left in the source.

Table 8: The fixed preprocessing contract, for the ODIM HDF5 product of Michelson et al. (2014) distributed through Met Office (2026). The three dBR constants are the ones that matter most, because the log transform is applied before optical flow, before the codec and before the diffusion target is formed.

Group	Setting	Value
Source grid	product	Met Office UK radar rainfall composite, ODIM HDF5
	size	2175×1725 pixels at 1 km, sampled every 15 min
	projection	transverse Mercator on the British National Grid
	record used	2024-11-21 to 2026-08-09, a rolling 19-month window
Three-valued logic	-1.0	nodata , masked to not-a-number, excluded everywhere
	0.0	undetected , genuinely dry, kept
	clip	rates above 128 mm h^{-1} , where the composite is not trusted
Log transform	wet threshold	DBR_THRESH = 0.1 mm h^{-1}
	dry floor	DBR_ZERO = -15.0 dBR
	transform	$\text{dB}(R) = 10 \log_{10} R$ on wet pixels only
Crop geometry	scored crop	256×256 pixels, so 256 km square at 1 km
	context	384×384 , advected then centre-cropped to 256
	margin	64 pixels, one hour of a 15 m s^{-1} storm
	stride	256, so candidate tiles do not overlap
Quality filter	candidates	42 tile positions, of which 12 ever pass
	in range	at least 90 percent of the crop's pixels valid
Samples	wet	at least 5 percent of them at or above 0.1 mm h^{-1}
	history	4 consecutive frames, 45 minutes of past
Leads	leads	+15, +30, +45, +60 min, one lead-conditioned model

The filter is the single most consequential preprocessing decision in the project, because it sets the base rate every score is computed against. It keeps 260,492 unique crops, cached as 1,041,968 lead-conditioned files, four per crop. Those 260,492 split into 153,473 train, 13,281 validation and 93,738 test. The twelve surviving tiles do not contribute equally. The busiest, centred on north-west England and the Irish Sea, passes 31,761 times, and the quietest passes 15,984, a spread of about two to one. Figure 7 shows where the accepted crops sit. The consequence, stated once here and referred to in Section 7.3, is that the evaluation population is wetter and better observed than the national grid as a whole, so absolute scores in this report sit above what a full-composite study would report, and only the within-report comparisons are like-for-like.

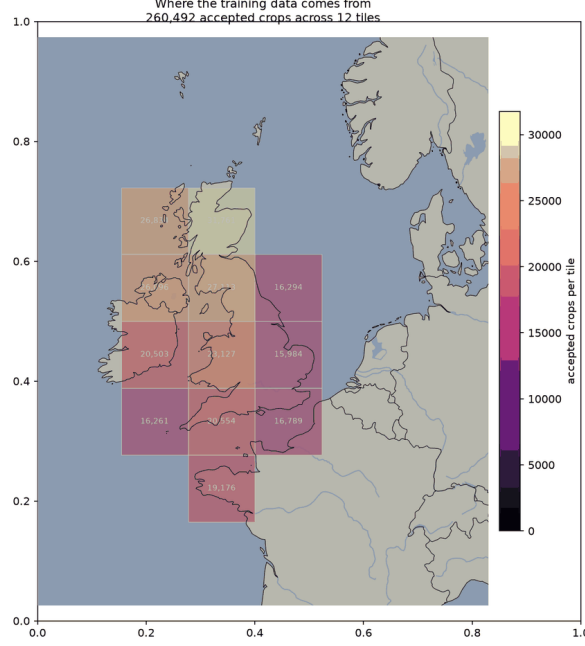


Figure 7: How often each accepted tile passes the quality filter over the whole record. Coverage concentrates over the west and the Irish Sea, where the radar network is dense and frontal rain is frequent, and thins over the far north and the south-east.

B Model and training configuration

Component	Setting	Value
Codec (VAE)	architecture	2 residual blocks per resolution, self-attention
	latent	$4 \times 64 \times 64$ from 256×256 (fourfold spatial)
	losses	DGMR-weighted masked $L_1 + 10^{-6}$ KL + hinge patch adversarial
	parameters	3.54 M generator, 2.76 M discriminator
	selected	epoch 17 of 25, val weighted L_1 0.0331, PSD ratio 0.957
Diffusion (both arms)	U-Net	width 128, mults (1,2,4), attention at 16, 65.7 M params
	conditioning	20 channels (Arm A) or 24 (Arm B), dropout 0.1
	EDM	$P_{mean} = -1.2$, $P_{std} = 1.2$, σ_d measured per arm
	σ_d	0.7407 (Arm A, target δ), 0.5341 (Arm B, target r')
	latent scale	0.78415, measured on the training pack
	optimiser	AdamW, lr 10^{-4} , cosine after 1000 warmup steps
	EMA / clip / precision	0.999 / 1.0 / bf16 autocast
Regression stage (Arm B)	epochs	50, scored at epoch 25 by the precommitted rule
	architecture	identical U-Net with the noise input pinned, 65.7 M
	loss / epochs	plain MSE on δ , 8 epochs, best at epoch 2
Sampler	explained variance	0.461 validation, 0.425 to 0.477 by lead
	scheme	deterministic Heun, 25 steps, $\rho = 7$, $\sigma \in [0.002, 80]$
Compute	ensemble	8 members from 8 independent seeds, guidance 1.0, churn 0
	hardware	JASMIN Orchid A100 (training), Exeter L4 (development)
	per arm	26.2 GPU-hours (Arm A) and 26.3 (Arm B), 50 epochs each
	throughput	about 325 samples per second, 31.5 minutes per epoch
memory		17.8 GB peak at batch 64 in bf16

Table 9: Full configuration of every learned stage. Both arms were run from the same seed, on the same packed latents, with the same dataloader, so the only intended difference between them is the diffusion target and the four extra conditioning channels that carry μ_r .

C The codec, what it reconstructs and what it costs

No result in this report can be read without knowing what the frozen autoencoder alone permits, because every model output is decoded through it. Both bounds were measured on the full validation split rather than estimated from a sample.

Table 10: Codec bounds on the validation split. The ceiling is the observation y put through an encode-decode round trip, so it is the best any latent-space model here could possibly score. The floor is the advection prior through the same round trip, so the difference between it and raw advection is the price of working in latent space at all.

Quantity	MAE ↓	RMSE ↓	Bias	CSI@1 ↑	CSI@8 ↑	Wet area
Observation (definitionally perfect)	0	0	0	1	1	0.2305
Ceiling, y through the codec	0.0428	0.1940	+0.0071	0.8872	0.7378	0.2297
Raw advection prior	0.2656	1.0177	−0.0164	0.4197	0.0861	0.2199
Floor, A through the codec	0.2672	n/a	n/a	n/a	n/a	n/a

The round trip costs 0.0016 mm h^{-1} , which is 0.6 percent of the advection baseline’s error. The ceiling’s 2 to 8 km spectral power ratio is 0.904, 0.895, 0.908 and 0.904 at the four leads, a mean of 0.903, and that is the number the checkpoint rule’s tiebreak targets. The important consequence is stated in Section 5.3: a member whose power ratio exceeds 0.903 is not sharper than the observation, it is sharper than the decoder can faithfully represent, so the excess is synthesised rather than recovered.

Figure 8 is the check that mattered when choosing this codec over the first one. The first version reconstructed accurately on average and still lost more than half the spatial power below 8 km, which would have capped the whole pipeline below its own baseline at exactly the scales the project is about. The replacement holds the intensity distribution across four orders of magnitude and tracks the observed spectrum down to the grid scale, with the only visible departure in the last two wavenumbers.

Two details behind the epoch-17 selection of Section 4.2 belong here rather than in the main text. Training ran 25 epochs with early stopping on weighted validation reconstruction loss, and the adversarial term used a two-epoch warmup before the adaptive weight took over. That weight rescales the adversarial gradient by the ratio of the reconstruction gradient norm to the adversarial gradient norm, both taken at the last decoder layer, so the two terms arrive at that layer with comparable magnitude no matter how the discriminator is doing. Epoch 20 reached a 2 to 8 km power ratio of 1.004, closer to unity than epoch 17’s 0.957, and was still rejected, because it is the first epoch at which the discriminator’s real logit collapses, from 0.730 at the selected epoch 17 and 0.603 at epoch 19 to 0.248 at epoch 20, and the latent standard deviation begins a monotone decline, from 1.284 at epoch 17 to 1.042 at epoch 25. A power ratio bought by a collapsing discriminator is not a sharper codec, it is a codec whose latent scale is drifting, and the diffusion stage downstream depends on that scale being stable.

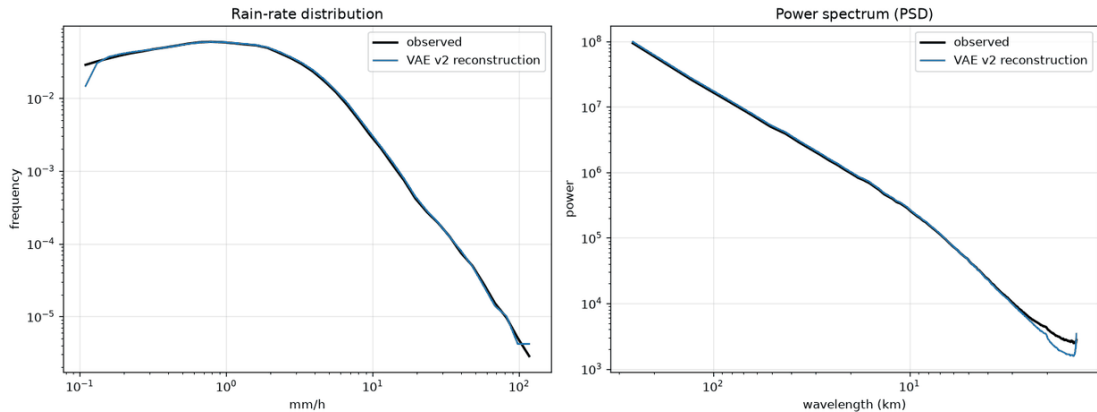


Figure 8: The selected codec at epoch 17, checked on held-out crops. Left, the rain-rate distribution over wet pixels. Right, the radially averaged power spectrum. The reconstruction follows the observation from 0.1 to above 100 mm h^{-1} and from 200 km down to about 3 km, which is the property the first codec did not have.

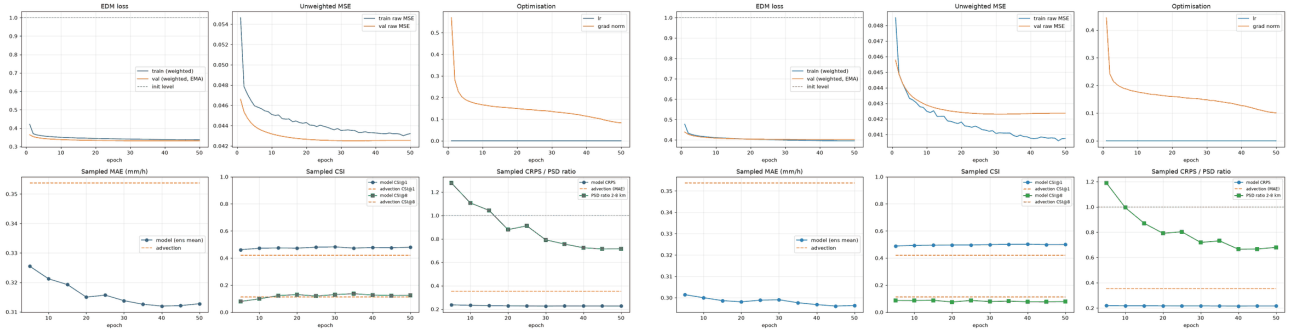
D Checkpoint selection decision table

The rule, fixed before either arm was scored: gate on sampled ensemble-mean error at or below the advection control on the same crops, then maximise the continuous ranked probability skill score (CRPSS), then break ties within 0.005 CRPSS on member spectral power closest to the codec ceiling of 0.903. All ten archived checkpoints of each arm pass the gate. Both arms independently select epoch 25.

Arm A (direct)				Arm B (two-stage)			
Epoch	CRPSS	Member PSD	Tied set	Epoch	CRPSS	Member PSD	Tied set
5	0.33044	1.2780		5	0.38032	1.1901	
10	0.34222	1.1076		10	0.38452	0.9949	
15	0.34862	1.0434		15	0.38473	0.8711	
20	0.35405	0.8790		20	0.38693	0.7913	yes
25	0.35743	0.9115	selected	25	0.38805	0.8033	selected
30	0.35908	0.7920	yes	30	0.38764	0.7187	yes
35	0.35716	0.7561	yes	35	0.39024	0.7315	yes
40	0.35836	0.7242	yes	40	0.39110	0.6653	yes
45	0.35821	0.7143	yes	45	0.38987	0.6668	yes
50	0.35874	0.7155	yes	50	0.38999	0.6799	yes

Table 11: Reproduced mechanically from each arm’s training log, with no GPU and no recomputation involved. Within each tied set the winner is the checkpoint whose member power ratio is closest to 0.903. The sensitivity noted in Section 7.3 is visible here. Arm B’s epoch 15 sits 0.00137 outside the tolerance with a power distance of 0.0319 against the winner’s 0.0997, so a slightly looser tolerance would have chosen it.

The pattern the table records is the reason the rule exists. Member spectral power falls monotonically with epoch in both arms, from 1.28 to 0.72 in Arm A and from 1.19 to 0.68 in Arm B, while CRPSS keeps creeping up. Selecting on the probabilistic score alone would therefore have picked one of the last checkpoints in each arm, which are the smoothest fields either run produced. Figure 9 shows the same trade as it happened.



(a) Arm A, direct residual diffusion.

(b) Arm B, two-stage.

Figure 9: Training curves for both arms, 50 epochs each. Top row: the weighted EDM loss, the unweighted mean squared error, and optimisation health. Bottom row: the in-training 16-crop sampled diagnostic. The bottom right panel carries the point of Section 4.6, since the green spectral power ratio decays past the codec ceiling while the blue probabilistic score stays flat, so the two criteria disagree about which epoch is best.

E Per-lead validation scorecards

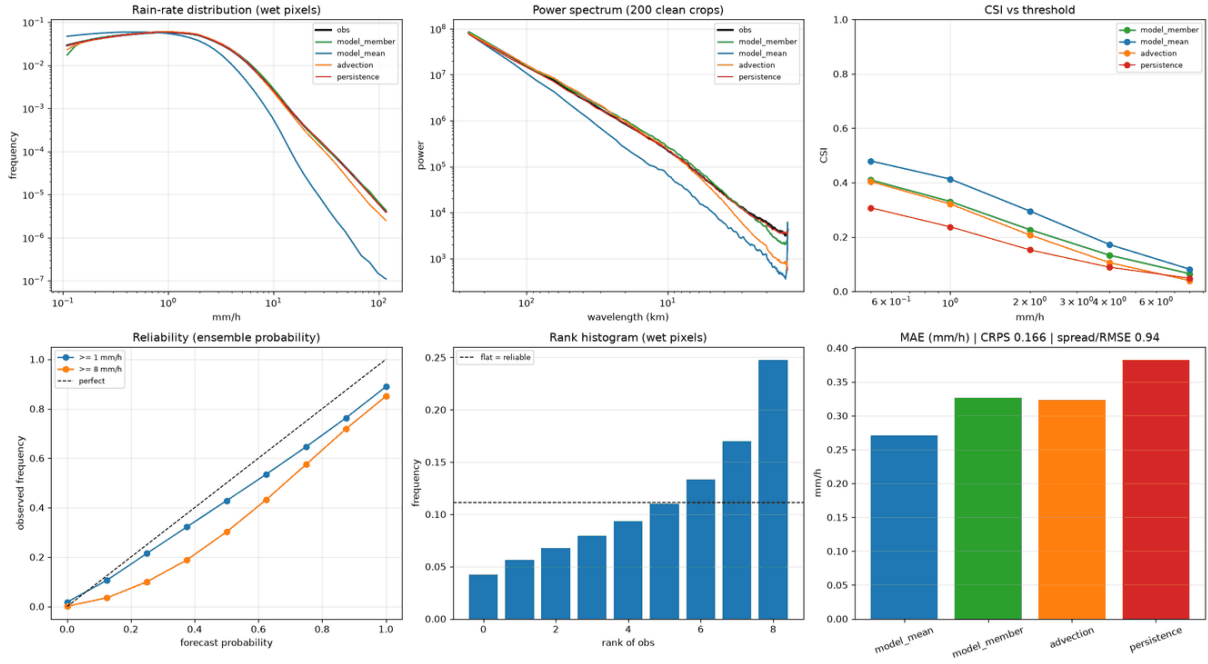
Lead	Output	MAE	CSI@1	CSI@8	Fair CRPS	Spread/RMSE	Member PSD	Mean PSD
+15	Advection	0.1931	0.5590	0.1681	n/a	n/a	0.487	n/a
	Regression mean	0.1516	0.6266	0.2366	n/a	n/a	n/a	0.132
	Arm A mean	0.1689	0.6149	0.2254	0.1068	0.9409	0.938	0.265
	Arm B mean	0.1655	0.6240	0.2348	0.1044	0.9420	0.997	0.284
+30	Advection	0.2529	0.4461	0.0881	n/a	n/a	0.511	n/a
	Regression mean	0.1886	0.5226	0.1441	n/a	n/a	n/a	0.102
	Arm A mean	0.2196	0.5138	0.1343	0.1359	0.9514	0.924	0.238
	Arm B mean	0.2136	0.5276	0.1446	0.1324	0.9528	1.069	0.266
+45	Advection	0.2934	0.3737	0.0566	n/a	n/a	0.547	n/a
	Regression mean	0.2124	0.4464	0.1036	n/a	n/a	n/a	0.087
	Arm A mean	0.2550	0.4440	0.0913	0.1562	0.9440	1.010	0.215
	Arm B mean	0.2466	0.4619	0.1038	0.1517	0.9469	1.021	0.239
+60	Advection	0.3231	0.3212	0.0398	n/a	n/a	0.559	n/a
	Regression mean	0.2294	0.3873	0.0803	n/a	n/a	n/a	0.074
	Arm A mean	0.2823	0.3909	0.0687	0.1718	0.9352	1.017	0.202
	Arm B mean	0.2709	0.4128	0.0818	0.1660	0.9383	0.996	0.226

Table 12: Validation split, 13,281 crops at each lead. Member PSD and mean PSD are the 2 to 8 km mean-of-ratios spectral power against observations. The codec ceiling by lead is 0.904, 0.895, 0.908 and 0.904. Arm B leads Arm A on error and on both categorical columns at every one of the four leads, which is why the pooled margin in Table 3 is not an artefact of pooling.

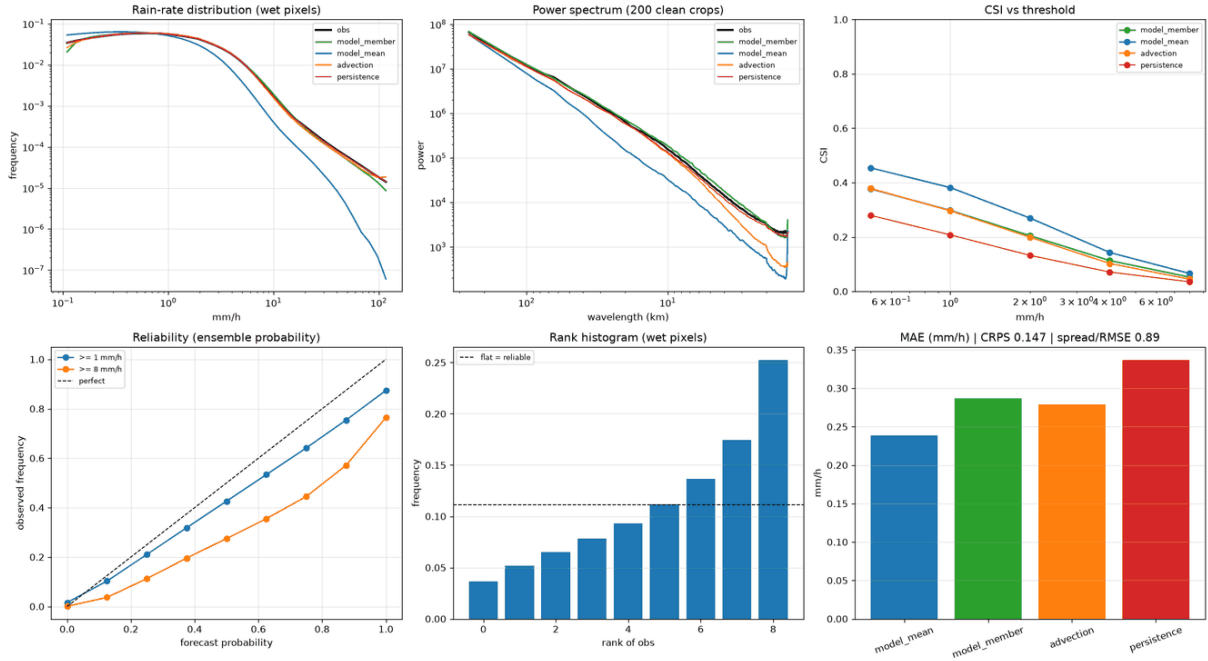
F Ensemble calibration diagnostics

Section 6.3 reports dispersion as two numbers, a spread ratio and an outlier rate. Those two numbers compress a lot, and Figure 10 shows what they compress. The top row carries the RQ3 trade directly: a single member follows the observed rain-rate distribution across four orders of magnitude and tracks the observed spectrum to the grid scale, while the ensemble mean of the very same eight members falls two orders of magnitude short in the heavy tail and loses power steadily below about 20 km. The mean is nonetheless the better forecast in the bottom right panel, which is the whole tension of the section stated in one figure.

The bottom middle panel is the diagnostic the pooled tables cannot show. A calibrated eight-member ensemble would give a flat rank histogram at 1/9 per bin. This one rises monotonically to the right and puts about 25 percent of wet pixels in the top rank, meaning the observation exceeded all eight members more than twice as often as it should. The under-dispersion is therefore not symmetric. It is concentrated on the high side, which is consistent with the reliability curve sitting below the diagonal at both thresholds, and it says that where these models do put rain they systematically put too little of it. That is the same defect the wet-area column of Table 2 records from the other direction, since the ensemble mean spreads light rain over too much area while under-forecasting intensity where rain actually falls. On the held-out year the shape is unchanged and the spread ratio falls from 0.94 to 0.89, which is the deterioration Section 6.5 attributes to error growing while spread does not.



(a) Validation split, 13,281 crops.



(b) Held-out 2026 test split, same checkpoint, same sampler.

Figure 10: Arm B at the rule-selected epoch 25, at +60 minutes. Top row: rain-rate distribution over wet pixels, radially averaged power spectrum, and critical success index against threshold. Bottom row: reliability of the ensemble exceedance probability, rank histogram over wet pixels with the flat calibrated line marked, and mean absolute error by method.

G The remaining four case frames

Section 6.6 shows two of the six held-out frames in full and summarises all six in Table 7. The other four are given here at the same size and with the same panel layout, because the argument of that section turns on reading across all six rather than on the two that happen to be printed in the main text. Each frame is a mosaic of the tiles that passed the quality filter at that timestamp, so the grey areas are out of radar range or below the filter, not dry.

The pair worth comparing directly is Figure 11 against Figure 13. On 19 June the advection prior is already accurate to 0.11 mm h^{-1} and the learned outputs have almost nothing left to correct, so the two ensemble means add wet area without adding skill and Arm A actually loses 1.5 percent. On 27 January the prior leaves 0.74 mm h^{-1} behind and the same two models recover 25 and 21 percent of it. That contrast is the boundary condition of the whole design, and it is invisible in every pooled table in this report. The other two frames fill in the middle of that range. Figure 12 is the one frame of the six where Arm A beats Arm B, and Figure 14 is the cleanest single illustration of the RQ3 trade, with the ensemble mean 29 percent too wet and its own members within 4 percent of the observed wet fraction.

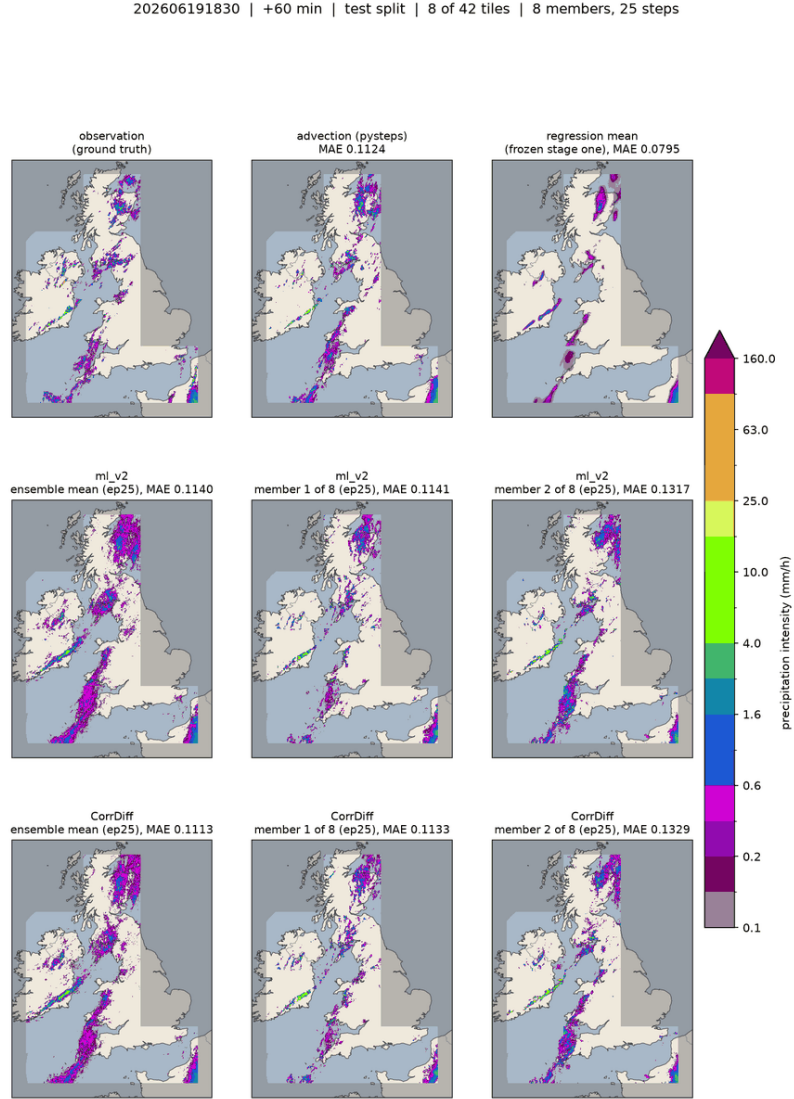


Figure 11: Convective, 19 June 2026 18:30 UTC, 8 tiles, +60 minutes. The lightest frame of the six, with an observed wet fraction of 0.105 and an advection error of 0.1124 mm h^{-1} . Arm B's ensemble mean gains 1.0 percent on the prior and Arm A loses 1.5. Both means put down more than 60 percent more wet area than was observed, which is the drizzle the diffusion contributes when there is no error to correct.

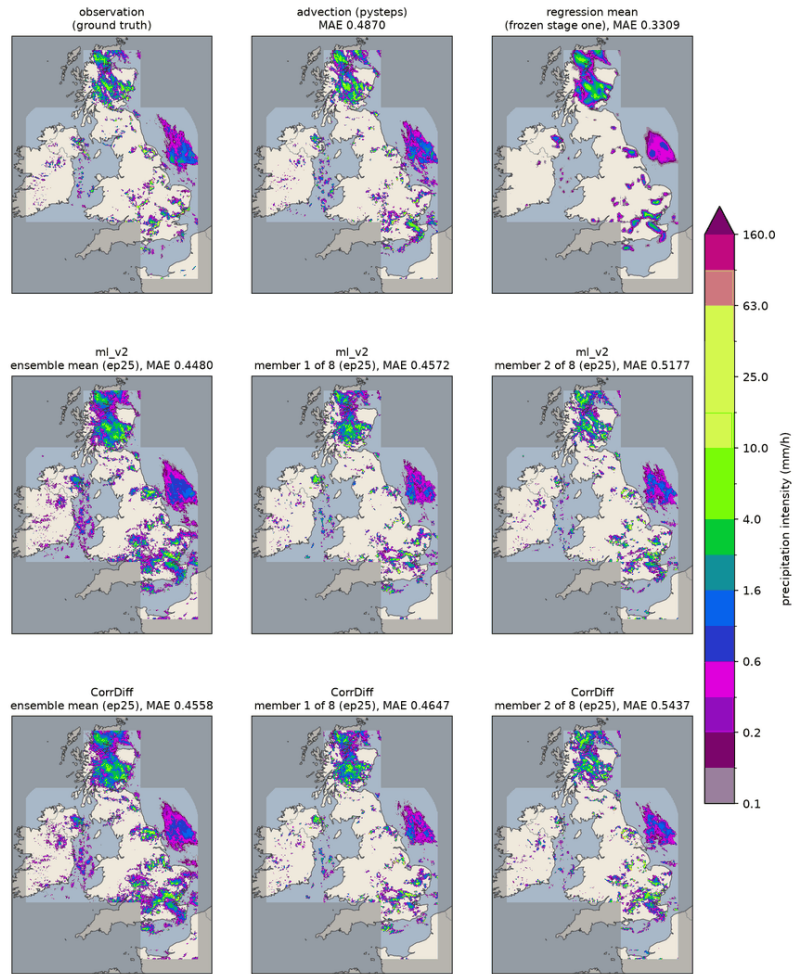


Figure 12: Convective, 2 June 2026 15:45 UTC, 8 tiles, +60 minutes. Advection leaves 0.4870 mm h^{-1} . This is the one frame of the six where Arm A beats Arm B, by 8.0 percent against 6.4, and it is reported because the pooled tables give Arm B the advantage without exception. The frozen regression mean is far ahead of both on error, at 32.1 percent, and is also the flattest field on the page.

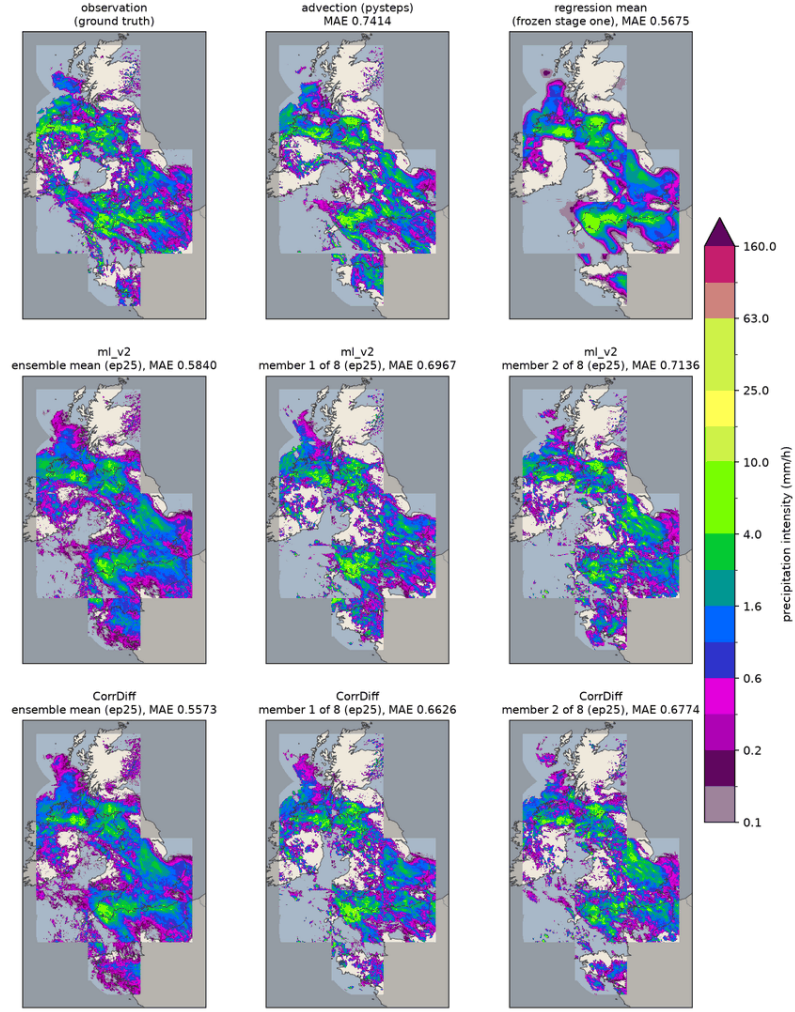


Figure 13: Frontal, 27 January 2026 05:00 UTC, 11 tiles, +60 minutes. The second heaviest frame, with an observed wet fraction of 0.579 and an advection error of 0.7414 mm h^{-1} . Arm B's ensemble mean gains 24.8 percent and a single member 10.6, the largest member gain anywhere in the six.

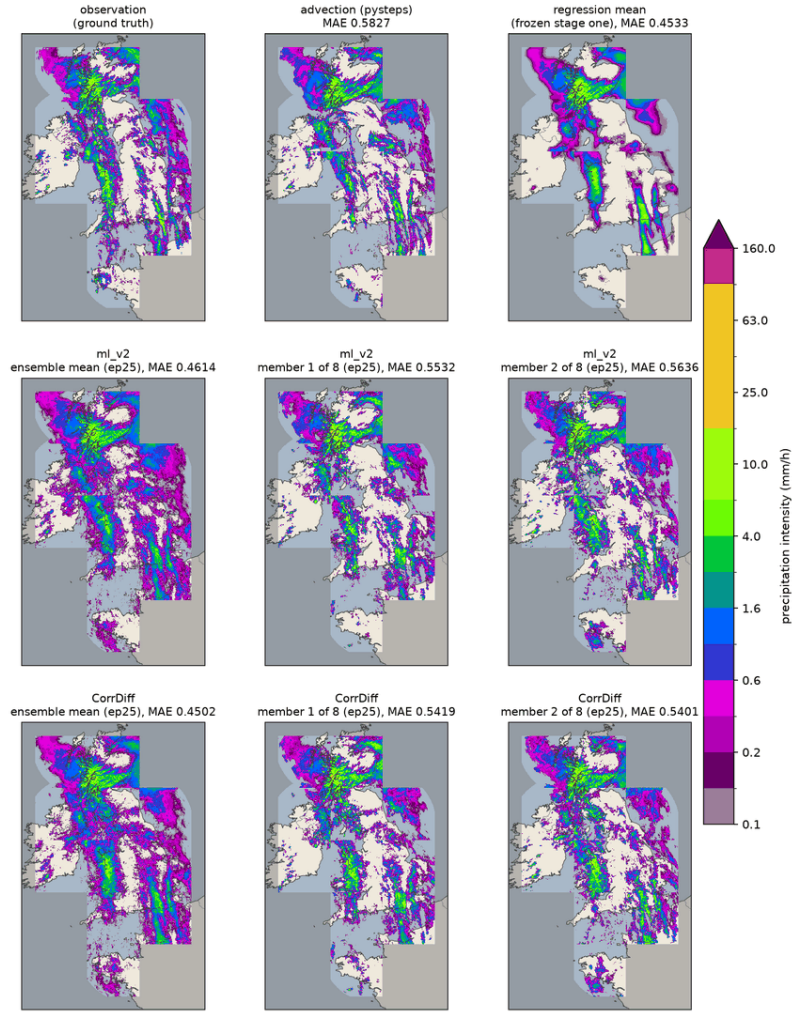


Figure 14: Frontal, 11 January 2026 19:15 UTC, 11 tiles, +60 minutes. Advection leaves 0.5827 mm h^{-1} . Arm B's ensemble mean gains 22.7 percent and its member 7.0. The member's wet area lands within 3.7 percent of observed where the ensemble mean is 29.3 percent too wet, which is the RQ3 trade visible on a single frame.

H Fractions skill score, full pooled table

Table 4 in the main text reduces the fractions skill score to one number per output, the smallest neighbourhood at which it is useful. That reduction hides how the two outputs get there, so the pooled curves behind it are tabulated here. Because the score is a ratio of sums, these are pooled from stored numerators and denominators rather than averaged across leads.

Table 13: Fractions skill score against neighbourhood width, pooled over all four leads on the validation split, for the two thresholds Table 4 reports. Pooled from 403 strided crops per lead, 1,612 in total, as stored numerators and denominators. Values above the usefulness target for that threshold are in bold. The targets are 0.547 at 1 mm h^{-1} and 0.501 at 8 mm h^{-1} , from observed base rates of 0.094 and 0.0022.

Threshold	Output	1 km	5 km	11 km	21 km	51 km	101 km
1 mm h^{-1}	Persistence	0.471	0.585	0.667	0.753	0.869	0.929
	Advection	0.593	0.727	0.802	0.861	0.926	0.959
	Arm B mean	0.673	0.793	0.853	0.898	0.945	0.968
	Arm B member	0.594	0.733	0.810	0.873	0.940	0.972
8 mm h^{-1}	Persistence	0.167	0.240	0.315	0.434	0.636	0.768
	Advection	0.189	0.357	0.492	0.612	0.754	0.829
	Arm B mean	0.311	0.477	0.578	0.673	0.777	0.815
	Arm B member	0.233	0.395	0.519	0.646	0.794	0.860

Two features of the heavy-rain block of Table 13 are worth naming. The ensemble mean leads the single member at every scale out to 21 km and then loses to it at 51 and 101, so the crossing referred to in Figure 5 happens between those two widths. And advection is beaten by both learned outputs at every scale in both blocks, with the single exception of the ensemble mean at 101 km and 8 mm h^{-1} , which is the fractions-skill restatement of RQ1.

Table 4 pools the two thresholds into a single question, the coarsest box each output needs before it is useful anywhere. Figure 15 plots that question against lead time for both arms. The two panels are nearly identical out to +45, and separate only at +60, where Arm B’s single member holds 21 km while its ensemble mean falls back to 51.

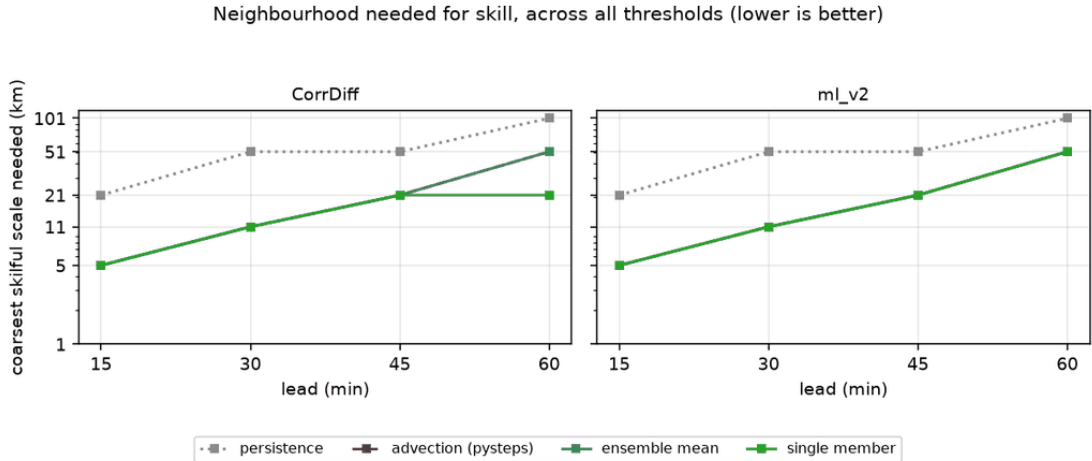


Figure 15: Coarsest neighbourhood needed for skill across all five thresholds, against lead time, for both arms. Lower is better. The advection line sits underneath the ensemble-mean line for most of both panels because the two need the same box at most leads, which is the RQ3 result seen from a third direction.

I Hyperparameter search, full effects table

Three studies were run: a nine-cell sampler sweep on a frozen checkpoint, a regression screen of nine configurations, and a U-Net screen of 25, the two training screens ladderred by successive halving with $\eta = 3$ over rungs of increasing data and epochs (Li et al., 2018). In total 43 distinct configurations ran as 56 trial-runs for about 22 GPU-hours, against a naive full-fidelity equivalent of 967 A100-hours for the two training screens alone. Every trial was paired on seed and batch.

The verdict is that the incumbent configuration was already near-optimal and nothing found is worth a retrain. That is a result rather than a failure, and it converts “the defaults were used” into “43 configurations were screened across nine parameters and the defaults survived”. The one-factor effects are in Table 14.

Table 14: Hyperparameter search, one-factor-at-a-time effects against a shared incumbent, paired on seed. The U-Net and regression columns are changes in each screen’s composite objective at the cheapest fidelity rung, and the sampler column is the inference objective on a frozen checkpoint. Positive is better.

U-Net screen (25 configurations)			Sampler sweep and regression screen		
Parameter	Value	Δ	Parameter	Value	Δ
width	192	+0.0287	steps (sampler)	18 (from 25)	+0.0130
P_{mean}	-0.81	+0.0018	members (sampler)	16 (from 8)	-0.0020
weight decay	10^{-2}	+0.0006	churn (sampler)	10 (from 0)	-0.0026
P_{std}	1.6	-0.0049	guidance (sampler)	1.25 (from 1.0)	-0.0336
width	96	-0.0116	guidance (sampler)	1.5 (from 1.0)	-0.0678
P_{mean}	-1.6	-0.0141	lr (regression)	2×10^{-4}	+0.0058
dropout	0.05	-0.0272	width (regression)	192	+0.0043
attention	16,32	-0.0302	dropout (regression)	0.05	-0.0053
dropout	0.10	-0.0317	lr (regression)	5×10^{-5}	-0.0151

Two further findings sit behind the table. Ensemble size is settled by a subtle observation the table does not show. The probabilistic score is flat from 4 to 16 members, while the spread ratio measured against its own member-dependent ideal holds at 0.974, 0.971 and 0.968 at 4, 8 and 16 members. The under-dispersion is therefore intrinsic to the model, and no ensemble size fixes it. Eighteen sampler steps is a real efficiency finding, matching the probabilistic score of 25 steps for 25 percent less compute. It was not adopted, because its member power ratio overshoots the codec ceiling by 9 percent and changing the step count would unpair every scorecard already produced. Churn buys the best raw probabilistic score of any sampler cell and near-ideal dispersion, but it overshoots to 0.979 against the 0.9428 ideal and pushes member power to 130 percent of the codec ceiling, so the objective ranks it below the incumbent for exactly the reason Section 5.3 gives.

J Reproducing the results

The pipeline is a sequence of stage scripts with file-level handoffs, not an application. Each stage skips outputs that already exist, and every checkpoint and JSON write goes through a temporary file followed by an atomic rename, so a killed job resumes rather than corrupting its own inputs. Table 15 lists the scripts in the order they run and what each one produces.

Table 15: The pipeline, stage by stage. Times are for the hardware in Appendix B. Stage 1 is the only CPU-bound stage and the only one that touches the network.

Stage	Script	Produces
1. Prior	<code>build_prior.py</code>	one .npz per accepted crop
	<code>clean_cache.py</code>	removes truncated .npz
	<code>evaluate_prior.py</code>	the baseline scorecard
	<code>fss.py, residuals.py</code>	scale and residual analysis
	<code>plot_domain.py</code>	Figure 1, the tile map
2. Codec	<code>pack_fields.py</code>	float16 memmaps of y and A
	<code>train_vae.py</code>	the codec, 25 epochs
	<code>train_vae_v1.py, compare_vae.py</code>	the superseded codec and the v1 to v2 comparison
3. Diffusion	<code>pack_latents.py</code>	cached latents, all crops and leads
	<code>ridge_gate.py</code>	the linear lower bound of Section 4.4
	<code>train_regression.py</code>	the frozen mean stage R_ϕ
	<code>pack_mu.py</code>	cached μ_r for Arm B
	<code>train_ldm.py, train_corrdiff.py</code>	Arm A and Arm B
4. Scoring	<code>select_ckpt.py</code>	the Appendix D decision
	<code>sample.py, evaluate.py</code>	the per-lead scorecards
	<code>evaluate_mean.py</code>	the regression-mean scorecard
	<code>pool_leads.py</code>	the pooled scorecards
	<code>plot_fss.py, plot_case.py</code>	the Section 6 and 7 figures
5. Search	<code>hpo_spaces.py, hpo_search.py</code>	the three screens and their fidelity ladders
	<code>hpo_baselines.py, hpo_report.py</code>	the shared incumbent and Table 14